Brazilian E-Commerce Analytics Platform

End-to-End Data Engineering, Analytics & Business Intelligence Project

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Technologies: Python, Pandas, MySQL, SQL, Power BI, AWS,

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Project Documentation

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1. Executive Summary

The Brazilian E-Commerce Analytics Platform is an end-to-end Data Engineering and Business Intelligence project designed to transform raw e-commerce data into meaningful business insights. Using the Brazilian E-Commerce Public Dataset, the project processes data related to customers, orders, products, sellers, payments, reviews, and geolocation information.

The solution follows a complete data engineering workflow, including data extraction, cleaning, transformation, and loading (ETL) using Python and Pandas. The transformed data is organized into a Star Schema data warehouse in MySQL, consisting of fact and dimension tables optimized for analytical reporting.

To support business decision-making, multiple SQL analytical views are created to analyze revenue, customer behavior, product performance, seller performance, payment trends, review ratings, and logistics efficiency. These analytical datasets are further visualized through interactive Power BI dashboards that provide real-time insights into key business metrics and performance indicators.

The project demonstrates industry-standard practices in data engineering, data warehousing, business intelligence, and cloud deployment. By integrating Python, MySQL, SQL, Power BI, and AWS, the platform delivers a scalable and efficient analytics solution that helps stakeholders monitor performance, identify trends, and make data-driven decisions.

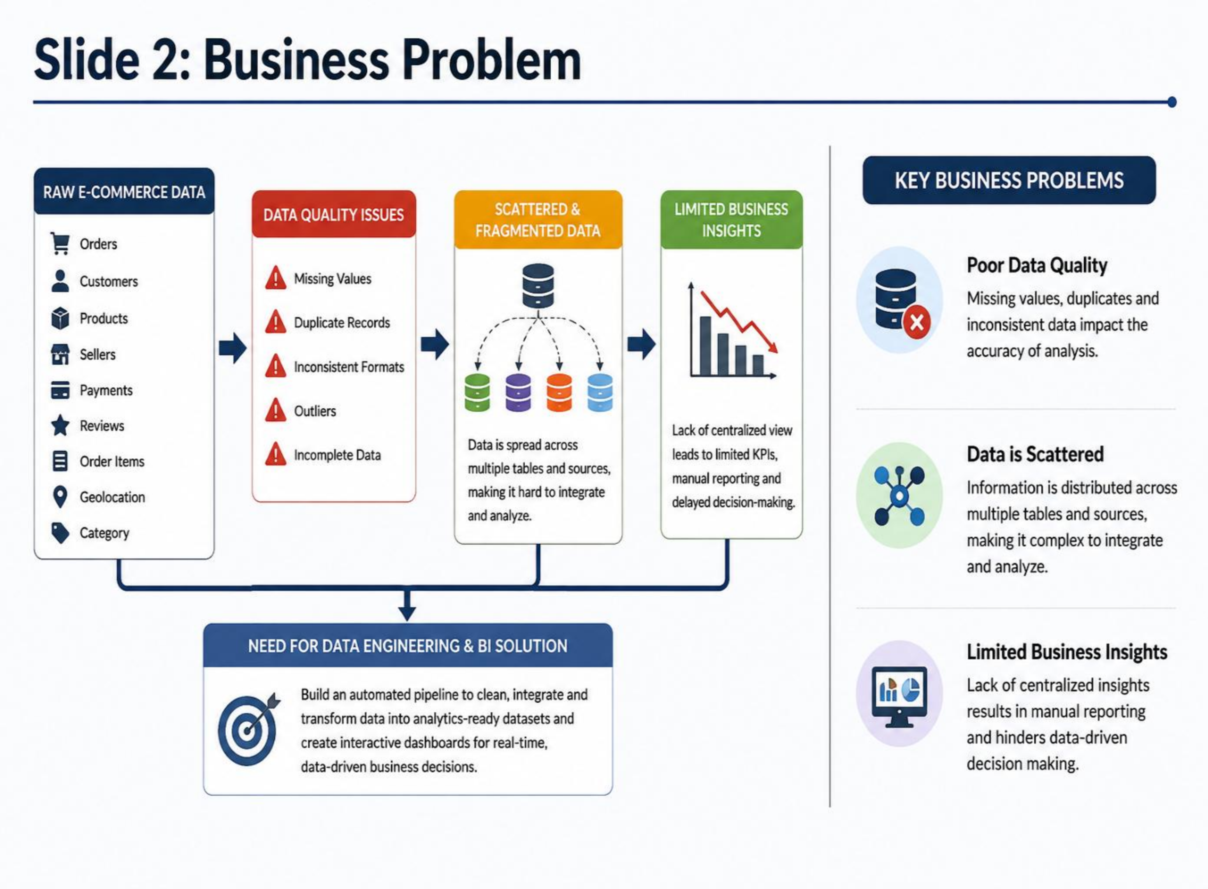
Overall, this project showcases the complete lifecycle of a modern analytics platform, from raw data processing to executive-level reporting, making it a practical real-world implementation of Data Engineering and Business Intelligence concepts.

2. Project Overview

The Brazilian E-Commerce Analytics Platform is an end-to-end Data Engineering and Business Intelligence project developed using the Brazilian E-Commerce Public Dataset. The project focuses on transforming raw transactional data into meaningful business insights through a structured data engineering pipeline, dimensional data modeling, SQL analytics, and interactive Power BI dashboards.

The primary objective of this project is to design and implement a scalable analytics solution that enables stakeholders to monitor sales performance, customer behavior, product trends, seller performance, payment patterns, and logistics efficiency. The project demonstrates industry-standard data engineering practices including data extraction, transformation, loading (ETL), data warehousing, business intelligence reporting, and cloud deployment.

3. Business Problem Statement



Expected Benefits

- Faster and more accurate reporting.
- Improved visibility into business operations.
- Better understanding of customer behavior.
- Enhanced product and seller performance analysis.
- Data-driven strategic decision-making.
- Increased operational efficiency and business growth.

4. Project Objectives

Primary Objective

The primary objective of this project is to design and develop an end-to-end E-Commerce Analytics Platform that transforms raw transactional data into actionable business insights through data engineering, data warehousing, analytics, and business intelligence solutions.

Specific Objectives

1. Data Integration

- Collect and integrate data from multiple e-commerce datasets, including customers, orders, products, sellers, payments, reviews, and geolocation data.
- Create a centralized repository for analytical processing.

2. Data Cleaning and Transformation

- Identify and handle missing, duplicate, and inconsistent records.
- Standardize and transform raw data into a clean and structured format suitable for analysis.
- Ensure data quality and consistency across all datasets.

3. ETL Pipeline Development

- Develop automated ETL (Extract, Transform, Load) processes using Python and Pandas.
- Convert raw datasets into analytics-ready tables.
- Improve data processing efficiency and reliability.

4. Data Warehouse Design

- Design and implement a Star Schema data warehouse model.
- Create fact and dimension tables to support analytical reporting.
- Optimize data storage and query performance.

5. Business Analytics Implementation

- Develop SQL-based analytical views to answer key business questions.
- Analyze revenue, customer behavior, product performance, seller performance, payment trends, and customer satisfaction.
- Generate meaningful business metrics and KPIs.

6. Dashboard Development

- Create interactive Power BI dashboards for business users and stakeholders.
- Visualize key performance indicators through charts, graphs, maps, and reports.
- Enable real-time monitoring of business performance.

7. Performance Measurement

- Track critical business KPIs such as:
    - Total Revenue
    - Total Orders
    - Total Customers
    - Average Order Value
    - Customer Satisfaction Score
    - Delivery Performance Metrics

8. Cloud Deployment

- Deploy the analytics solution on AWS cloud infrastructure.
- Ensure scalability, reliability, and accessibility of the platform.
- Demonstrate cloud-based data engineering practices.

9. Business Insight Generation

- Identify trends, patterns, and opportunities within e-commerce operations.
- Support data-driven decision-making through analytical reporting.
- Provide recommendations for business improvement.

10. Industry-Level Data Engineering Experience

- Apply real-world data engineering concepts and best practices.
- Gain hands-on experience in ETL development, data modeling, analytics, visualization, and cloud deployment.
- Build a portfolio-ready project that reflects industry standards.

Project Outcome

By achieving these objectives, the project delivers a complete Business Intelligence and Analytics solution that enables stakeholders to efficiently analyze e-commerce operations, monitor performance, and make informed strategic decisions based on data.

5. Technology Stack

Component	Technology
Programming Language	Python
Data Processing	Pandas
Database	MySQL
Analytics	SQL
Business Intelligence	Power BI
Cloud Platform	AWS
Version Control	Git & GitHub
Workflow Orchestration (Future)	Apache Airflow

Key Deliverables

- Data Cleaning and Transformation Pipeline
- Dimensional Data Warehouse
- Fact and Dimension Tables
- Business Analytics Views
- Power BI Interactive Dashboards
- AWS Deployment Architecture
- Business Insights and Recommendations

Expected Business Value

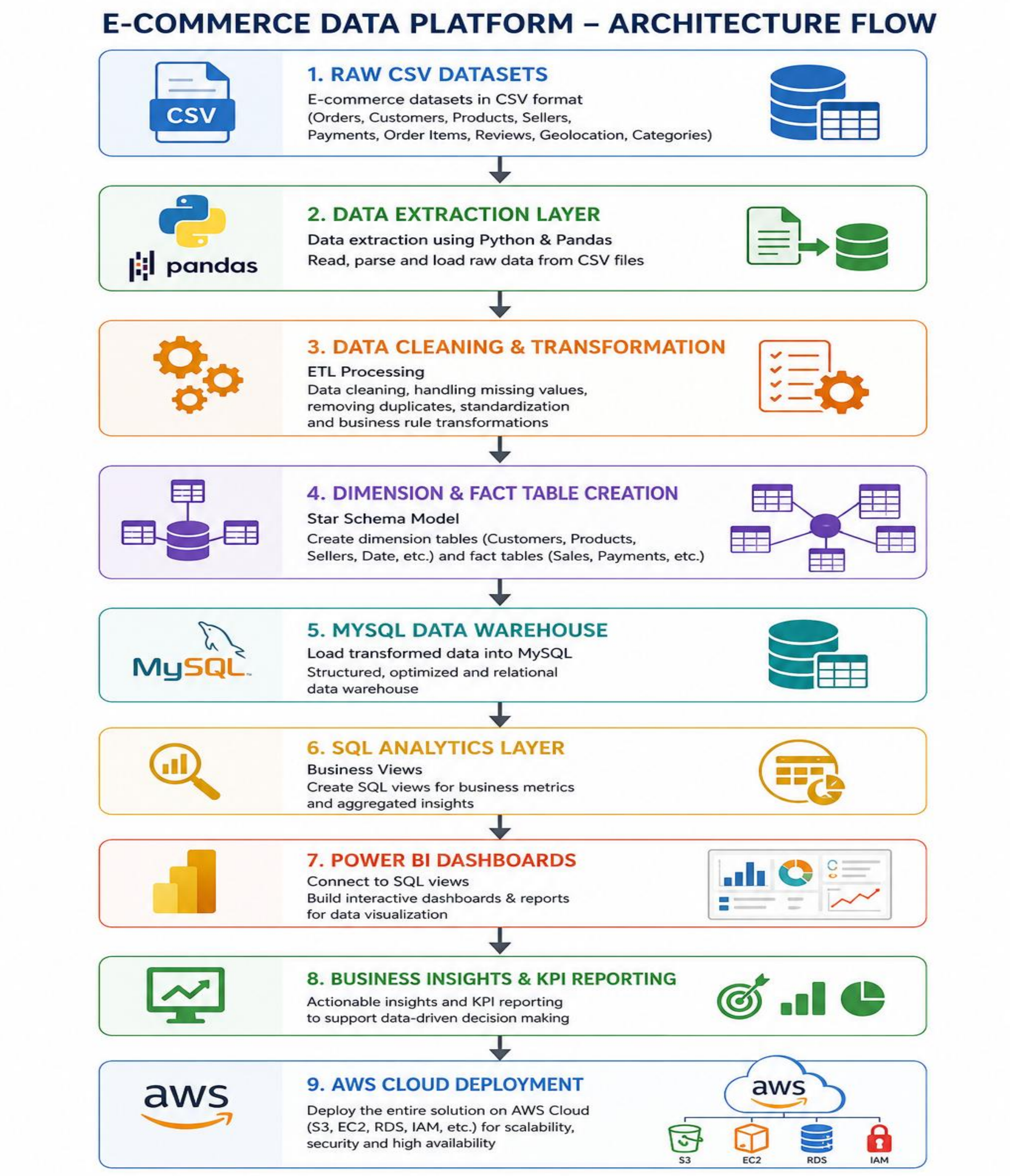
The analytics platform enables business users to:

- Monitor overall sales performance.
- Identify top-performing products and sellers.
- Analyze customer purchasing behavior.
- Track payment method preferences.
- Evaluate delivery performance and logistics efficiency.
- Measure customer satisfaction through review analysis.
- Support strategic business decision-making through data-driven insights.

6. System Architecture

The Brazilian E-Commerce Analytics Platform follows a modern Data Engineering and Business Intelligence architecture that transforms raw e-commerce data into actionable business insights. The system consists of multiple layers including data ingestion, data processing, data warehousing, analytics, visualization, and cloud deployment.

The architecture ensures efficient data flow, scalability, and support for analytical reporting and business decision-making.



7. Dataset Description

Overview

The Brazilian E-Commerce Public Dataset contains transactional and operational data from an e-commerce marketplace in Brazil. The dataset includes information about customers, orders, products, sellers, payments, reviews, order items, and geolocation data. These datasets serve as the foundation for building the data warehouse and analytics platform.

7.1 Customers Dataset

Description

The Customers Dataset contains information about customers who placed orders on the e-commerce platform.

Column Name	Description
customer_id	Unique identifier for each customer
customer_unique_id	Unique identifier representing a customer across multiple orders
customer_zip_code_prefix	Customer ZIP code prefix
customer_city	Customer city
customer_state	Customer state

Purpose

- Customer segmentation
- Geographic analysis
- Customer behavior analysis
- Customer distribution reporting

7.2 Orders Dataset

Description

The Orders Dataset contains information about customer orders and their lifecycle status.

Column Name	Description
order_id	Unique order identifier
customer_id	Customer identifier
order_status	Current order status
order_purchase_timestamp	Order purchase date and time
order_approved_at	Order approval timestamp
order_delivered_carrier_date	Carrier pickup date
order_delivered_customer_date	Customer delivery date
order_estimated_delivery_date	Estimated delivery date

Purpose

- Order tracking
- Sales trend analysis
- Delivery performance analysis
- Revenue reporting

7.3 Products Dataset

Description

The Products Dataset contains details about products sold on the platform

Key Columns

Column Name	Description
product_id	Unique product identifier

Column Name	Description
product_category_name	Product category
product_name_length	Product name length
product_description_length	Product description length
product_photos_qty	Number of product photos
product_weight_g	Product weight in grams
product_length_cm	Product length
product_height_cm	Product height
product_width_cm	Product width

Purpose

- Product performance analysis
- Inventory insights
- Product category reporting
- Product characteristics analysis

7.4 Product Category Dataset

Description

The Product Category Dataset provides translation of product category names from Portuguese to English.

Key Columns

Column Name	Description
product_category_name	Original category name (Portuguese)
product_category_name_english	Category name in English

Purpose

- Category standardization
- Reporting consistency
- Dashboard readability
- Product categorization analysis

7.5 Sellers Dataset

Description

The Sellers Dataset contains information about sellers operating on the marketplace.

Key Columns

Column Name	Description
seller_id	Unique seller identifier
seller_zip_code_prefix	Seller ZIP code prefix
seller_city	Seller city
seller_state	Seller state

Purpose

- Seller performance analysis
- Geographic seller distribution
- Revenue contribution analysis
- Marketplace performance evaluation

7.6 Payments Dataset

Description

The Payments Dataset contains transaction payment details for customer orders.

Key Columns

Column Name	Description
order_id	Order identifier
payment_sequential	Payment sequence number
payment_type	Payment method
payment_installments	Number of installments
payment_value	Payment amount

Purpose

- Revenue analysis
- Payment method analysis
- Installment behavior analysis
- Financial reporting

7.7 Reviews Dataset

Description

The Reviews Dataset contains customer ratings and feedback for completed orders.

Key Columns

Column Name	Description
review_id	Unique review identifier
order_id	Related order identifier
review_score	Customer rating score
review_comment_title	Review title
review_comment_message	Review description
review_creation_date	Review creation date
review_answer_timestamp	Review response timestamp

Purpose

- Customer satisfaction analysis
- Review trend analysis
- Sentiment evaluation
- Service quality measurement

7.8 Order Items Dataset

Description

The Order Items Dataset contains detailed information about products purchased within each order.

Key Columns

Column Name	Description
order_id	Order identifier
order_item_id	Item identifier within an order
product_id	Product identifier
seller_id	Seller identifier
shipping_limit_date	Shipping deadline
price	Product price
freight_value	Shipping cost

Purpose

- Product sales analysis
- Revenue calculations
- Seller contribution analysis
- Shipping cost analysis

7.9 Geolocation Dataset

Description

The Geolocation Dataset contains geographic information based on ZIP code prefixes.

Key Columns

Column Name	Description
geolocation_zip_code_prefix	ZIP code prefix
geolocation_lat	Latitude
geolocation_lng	Longitude
geolocation_city	City
geolocation_state	State

Purpose

- Geographic mapping
- Regional sales analysis
- Customer location analysis

- Logistics and delivery optimization

Dataset Summary

Dataset	**Purpose**
Customers	Customer information and demographics
Orders	Order lifecycle and status tracking
Products	Product information and attributes
Product Category	Category translation and standardization
Sellers	Seller information and performance
Payments	Payment transactions and revenue
Reviews	Customer feedback and satisfaction
Order Items	Product-level sales transactions
Geolocation	Geographic and location analytics

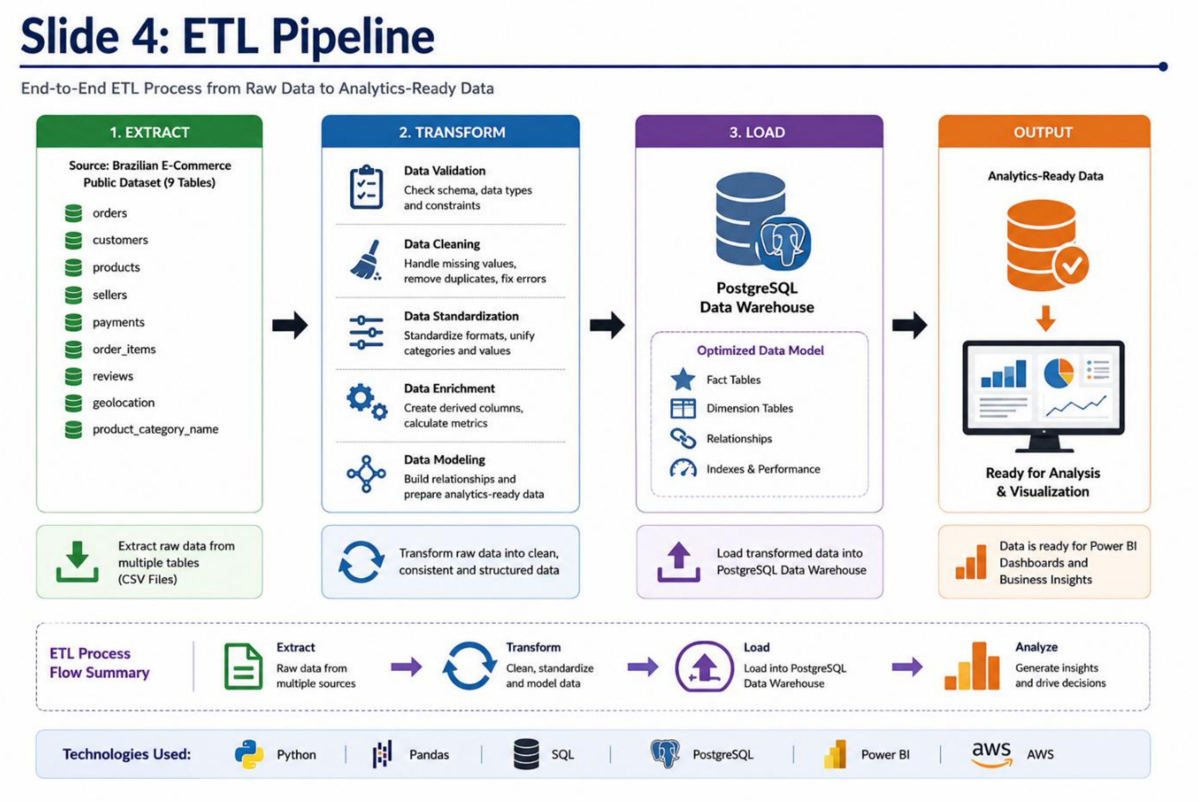
These datasets collectively provide a comprehensive view of the e-commerce business and serve as the foundation for ETL processing, data warehousing, SQL analytics, Power BI dashboards, and AWS deployment.

8. Data Engineering Pipeline

Overview

The Data Engineering Pipeline is the core component of the Brazilian E-Commerce Analytics Platform. It is responsible for extracting raw data, cleaning and transforming it into a structured format, and loading it into a data warehouse for analytical reporting and business intelligence.

The pipeline follows the ETL (Extract, Transform, Load) methodology and is implemented using Python and Pandas to ensure data quality, consistency, and scalability.



8.1 Data Extraction

Description

Data extraction is the first stage of the pipeline where raw datasets are collected and imported into the processing environment. The project uses multiple CSV files from the Brazilian E-Commerce Public Dataset.

Datasets Extracted

- Customers Dataset
- Orders Dataset
- Products Dataset
- Product Category Dataset

- Sellers Dataset
- Payments Dataset
- Reviews Dataset
- Order Items Dataset
- Geolocation Dataset

Tools Used

- Python
- Pandas

Activities Performed

- Reading CSV files using Pandas.
- Verifying dataset structure.
- Checking column names and data types.
- Performing initial data inspection.
- Measuring dataset size and record counts.

Example

```
import pandas as pd

customers = pd.read_csv('customers.csv')

orders = pd.read_csv('orders.csv')

products = pd.read_csv('products.csv')
```

Output

Raw datasets loaded into Python DataFrames for further processing.

8.2 Data Cleaning

Description

Data cleaning ensures that the extracted data is accurate, complete, and ready for analytical processing. This stage focuses on identifying and correcting data quality issues.

Data Quality Checks

Missing Values Handling

- Identify null values.
- Fill or remove missing records where appropriate.
- Ensure critical business fields are not empty.

Duplicate Record Removal

- Detect duplicate rows.
- Remove redundant records.

Data Consistency Validation

- Verify key relationships.
- Ensure valid identifiers across datasets.
- Validate date and numeric fields.

Data Type Standardization

- Convert date columns to datetime format.
- Standardize text formatting.
- Ensure numerical fields use correct data types.

Example Cleaning Activities

- Removed duplicate customer records.
- Standardized city and state names.
- Converted timestamp columns to datetime format.
- Validated product and seller identifiers.

Output

Clean and validated datasets ready for transformation.

8.3 Data Transformation

Description

Data transformation converts cleaned datasets into a business-friendly structure suitable for data warehousing and analytics.

Transformation Activities

Column Selection

- Retain only required business columns.
- Remove unnecessary attributes.

Feature Engineering

- Create surrogate keys.
- Generate analytical attributes.
- Create calculated fields.

Relationship Mapping

- Establish links between customers, orders, products, sellers, and payments.
- Prepare datasets for Star Schema implementation.

Date Processing

- Extract:
    - Year
    - Month
    - Quarter
    - Week
    - Day

Business Rule Implementation

- Revenue calculations.
- Delivery performance metrics.
- Customer behavior indicators.

Example Transformations

Customer Dimension

customer_id
customer_unique_id
customer_city
customer_state

Date Dimension

date_key
year
month
quarter
day
weekday

Payment Analysis Fields

payment_type
payment_value
payment_installments

Output

Analytics-ready dimension and fact datasets.

8.4 Data Loading

Description

The final stage of the pipeline loads transformed datasets into the MySQL Data Warehouse for reporting and business intelligence.

Data Warehouse Structure

Dimension Tables

- dim_customers
- dim_products
- dim_sellers
- dim_dates
- dim_geolocation

Fact Tables

- fact_orders
- fact_order_items
- fact_payments
- fact_reviews

Loading Process

Table Creation

- Create dimension tables.
- Create fact tables.
- Define primary and foreign key relationships.

Data Insertion

- Load transformed data into MySQL.
- Validate row counts.
- Verify data integrity.

Data Validation

- Check successful record insertion.
- Validate table relationships.
- Confirm warehouse consistency.

Technologies Used

- Python
- Pandas
- MySQL Connector
- MySQL Database

Output

A fully populated MySQL Data Warehouse optimized for analytical queries and reporting.

Pipeline Benefits

- Automated data processing workflow.
- Improved data quality and consistency.
- Faster analytical query performance.
- Scalable warehouse architecture.
- Reliable data source for Power BI dashboards.
- Support for business intelligence and decision-making.

9. Data Warehouse Design

Overview

The Brazilian E-Commerce Analytics Platform uses a **Star Schema Data Warehouse** to organize transactional data into fact and dimension tables for efficient analytics and reporting. The schema is designed to support business intelligence requirements such as revenue analysis, customer behavior analysis, seller performance evaluation, payment analysis, logistics monitoring, and customer satisfaction tracking.

The warehouse consists of **4 Fact Tables** and **6 Dimension Tables**, providing a scalable and optimized structure for SQL analytics and Power BI reporting.

9.1 Star Schema

Description

A Star Schema is a dimensional modeling technique where central fact tables store business transactions and are connected to surrounding dimension tables containing descriptive attributes.

Advantages of Star Schema

- Faster analytical queries
- Simplified reporting structure
- Easy integration with Power BI
- Improved dashboard performance
- Scalable data warehouse design

Star Schema Architecture

Based on the implemented data warehouse, the schema consists of:

Fact Tables

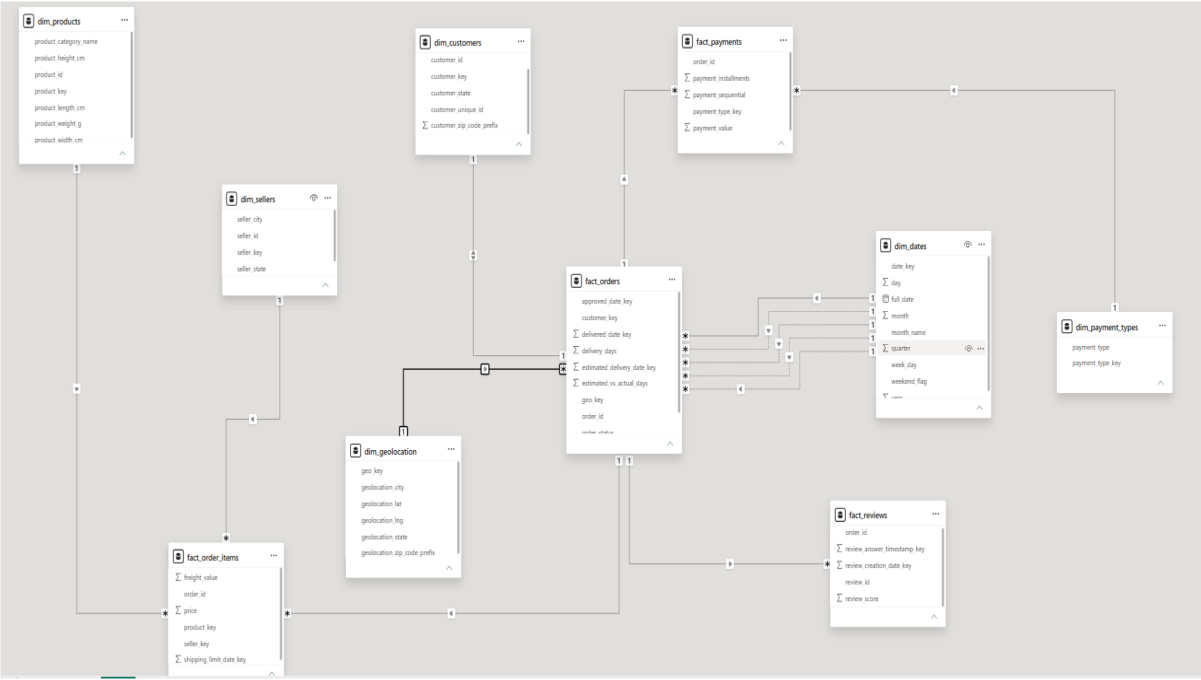
- fact_orders
- fact_order_items
- fact_payments
- fact_reviews

Dimension Tables

- dim_customers
- dim_products
- dim_sellers

- dim_dates
- dim_geolocation
- dim_payment_types

Relationship Overview



9.2 Dimension Tables

Dimension tables contain descriptive business information used for filtering, grouping, and analyzing facts

1. dim_customers

Description

The Customer Dimension stores customer demographic and geographic information. It provides customer-related attributes used for customer segmentation and regional analysis.

Primary Key

- customer_key

Columns

Column Name	Description
customer_key	Surrogate key for customer dimension

Column Name	Description
customer_id	Original customer identifier
customer_unique_id	Unique customer identifier across orders
customer_city	Customer city
customer_state	Customer state

Business Purpose

- Customer segmentation
- Regional customer analysis
- Customer purchasing behavior

2. dim_products

Description

The Product Dimension stores product information and characteristics sold through the e-commerce platform.

Primary Key

- product_key

Columns

Column Name	Description
product_key	Surrogate key for product dimension
product_id	Original product identifier
product_category_name	Product category
product_weight_g	Product weight in grams
product_length_cm	Product length
product_height_cm	Product height
product_width_cm	Product width

Business Purpose

- Product performance analysis
- Category-level reporting

3. dim_sellers

Description

The Seller Dimension stores information about marketplace sellers.

Primary Key

- seller_key

Columns

Column Name	Description
seller_key	Surrogate key for seller dimension
seller_id	Original seller identifier
seller_city	Seller city
seller_state	Seller state

Business Purpose

- Seller performance evaluation
- Regional seller analysis
- Revenue contribution reporting
-

4. dim_dates

Description

The Date Dimension provides time-related attributes for reporting and trend analysis.

Primary Key

- date_key

Columns

Column Name	Description
date_key	Surrogate key for date dimension
full_date	Calendar date
day	Day of month
month	Month number
quarter	Quarter number

Column Name	Description
year	Year
week_day	Day of week
weekend_flag	Weekend indicator

Business Purpose

- Time-series analysis
- Monthly and yearly trends
- Seasonal reporting

5. dim_geolocation

Description

The Geolocation Dimension stores location details mapped through ZIP code prefixes.

Primary Key

- geo_key

Columns

Column Name	Description
geo_key	Surrogate key for geolocation dimension
geolocation_zip_code_prefix	ZIP code prefix
geolocation_city	City name
geolocation_state	State name
geolocation_lat	Latitude
geolocation_lng	Longitude

Business Purpose

- Geographic analysis
- Regional revenue reporting
- Logistics optimization

6. dim_payment_types

Description

The Payment Type Dimension stores standardized payment methods used by customers.

Primary Key

- payment_type_key

Columns

Column Name	Description
payment_type_key	Surrogate key for payment type
payment_type	Payment method name

Business Purpose

- Payment trend analysis
- Financial reporting
- Customer payment preference analysis
-

9.3 Fact Tables

1. fact_orders

Description

The Orders Fact Table contains order-level transactional data and acts as the central fact table in the warehouse.

Primary Key

- order_key

Foreign Keys

- customer_key
- approved_date_key
- delivered_date_key
- estimated_delivery_date_key

Column Name	Description
order_key	Surrogate key for order
order_id	Original order identifier
customer_key	Customer foreign key
approved_date_key	Order approval date key
delivered_date_key	Delivery date key
estimated_delivery_date_key	Estimated delivery date key
order_status	Current order status
order_value	Total order value

Measures

- Total Orders
- Order Revenue
- Order Status Metrics

2. fact_order_items

Description

The Order Items Fact Table stores product-level sales transactions for each order.

Foreign Keys

- product_key
- seller_key
- shipping_limit_date_key

Columns

Column Name	Description
order_id	Order identifier
product_key	Product foreign key
seller_key	Seller foreign key
shipping_limit_date_key	Shipping date key

Column Name	Description
price	Product selling price
freight_value	Shipping cost

Measures

- Product Revenue
- Freight Charges
- Product Sales
-

3. fact_payments

Description

The Payments Fact Table stores all payment transactions associated with customer orders.

Foreign Keys

- payment_type_key

Columns

Column Name	Description
order_id	Order identifier
payment_type_key	Payment type foreign key
payment_sequential	Payment sequence number
payment_installments	Number of installments
payment_value	Payment amount

Measures

- Total Revenue
- Average Payment Value
- Installment Analysis

4. fact_reviews

Description

The Reviews Fact Table stores customer review ratings and feedback metrics.

Foreign Keys

- review_creation_date_key
- review_answer_timestamp_key

Columns

Column Name	Description
order_id	Order identifier
review_creation_date_key	Review creation date key
review_answer_timestamp_key	Review response date key
review_score	Customer rating score

Measures

- Average Review Score
- Customer Satisfaction Index
- Review Trends

Data Warehouse Statistics

Category	Count
Dimension Tables	6
Fact Tables	4
Total Tables	10
Schema Type	Star Schema
Database	MySQL
Reporting Tool	Power BI

10. ETL Process Documentation

Overview

The ETL (Extract, Transform, Load) process is responsible for converting raw Brazilian E-Commerce datasets into analytics-ready dimension and fact tables. Python and Pandas were used to automate the extraction, cleaning, transformation, and loading of data into the MySQL Data Warehouse.

10.1 Customer ETL

Objective

To create the **dim_customers** dimension table containing customer demographic and location information.

Source Dataset

- customers.csv

Extract

- Read customer data from CSV file.
- Validate column structure and record count.

Transform

- Remove duplicate customer records.
- Check and handle missing values.
- Standardize city and state names.
- Create surrogate key (customer_key).

Load

- Load transformed data into dim_customers.

Output Table

- dim_customers

Business Value

- Customer segmentation
- Geographic customer analysis
- Customer behavior reporting

10.2 Product ETL

Objective

To create the **dim_products** dimension table containing product information.

Source Dataset

- products.csv

Extract

- Import product dataset.

Transform

- Handle missing values.
- Validate product attributes.
- Standardize category names.
- Generate product_key.

Load

- Load data into dim_products.

Output Table

- dim_products

Business Value

- Product analysis
- Category reporting
- Product trend monitoring

10.3 Seller ETL

Objective

To create the **dim_sellers** dimension table.

Source Dataset

- sellers.csv

Extract

- Read seller data.

Transform

- Remove duplicates.
- Validate seller location fields.
- Create seller_key.

Load

- Insert data into dim_sellers.

Output Table

- dim_sellers

Business Value

- Seller performance analysis
- Regional seller reporting

10.4 Order ETL

Objective

To create the **fact_orders** fact table.

Source Dataset

- orders.csv

Extract

- Import order transactions.

Transform

- Convert timestamps to datetime format.
- Map customer_key from dim_customers.
- Generate order_key.
- Create date relationships.
- Validate order status values.

Load

- Load data into fact_orders.

Output Table

- fact_orders

Business Value

- Order tracking
- Revenue analysis
- Delivery performance reporting

10.5 Payment ETL

Objective

To create **fact_payments** and **dim_payment_types**.

Source Dataset

- order_payments.csv

Extract

- Load payment transaction data.

Transform

- Standardize payment types.
- Create payment_type_key.
- Validate payment values.
- Check installment information.

Load

- Load payment methods into dim_payment_types.
- Load payment transactions into fact_payments.

Output Tables

- dim_payment_types
- fact_payments

Business Value

- Revenue analysis
- Payment trend reporting
- Customer payment behavior analysis

10.6 Review ETL

Objective

To create the **fact_reviews** table.

Source Dataset

- order_reviews.csv

Extract

- Read review records.

Transform

- Handle missing review comments.
- Validate review scores.

- Convert review dates.
- Create review date relationships.

Load

- Load transformed data into fact_reviews.

Output Table

- fact_reviews

Business Value

- Customer satisfaction analysis
- Service quality measurement
- Review trend reporting

10.7 Product Category ETL

Objective

To standardize product categories and support product analytics.

Source Dataset

- product_category_name_translation.csv

Extract

- Load category translation dataset.

Transform

- Translate Portuguese categories to English.
- Remove duplicate category mappings.
- Validate category consistency.

Load

- Merge category information into dim_products.

Output

- Enhanced dim_products table

Business Value

- Standardized reporting
- Category-based sales analysis

10.8 Order Items ETL

Objective

To create the **fact_order_items** table.

Source Dataset

- order_items.csv

Extract

- Import order item transactions.

Transform

- Map product_key from dim_products.
- Map seller_key from dim_sellers.
- Convert shipping dates.
- Validate price and freight values.

Load

- Load transformed records into fact_order_items.

Output Table

- fact_order_items

Business Value

- Product sales analysis
- Seller contribution analysis
- Shipping cost analysis
-

10.9 Geolocation ETL

Objective

To create the **dim_geolocation** dimension table.

Source Dataset

- geolocation.csv

Extract

- Load geolocation data.

Transform

- Remove duplicate ZIP code entries.

- Validate latitude and longitude values.
- Standardize city and state names.
- Create geo_key.

Load

- Load transformed data into dim_geolocation.

Output Table

- dim_geolocation

Business Value

- Geographic analysis
- Regional sales reporting
- Logistics optimization

ETL Process Summary

ETL Process	Source Dataset	Output Table
Customer ETL	customers.csv	dim_customers
Product ETL	products.csv	dim_products
Seller ETL	sellers.csv	dim_sellers
Order ETL	orders.csv	fact_orders
Payment ETL	order_payments.csv	fact_payments, dim_payment_types
Review ETL	order_reviews.csv	fact_reviews
Product Category ETL	product_category_name_translation.csv	dim_products
Order Items ETL	order_items.csv	fact_order_items
Geolocation ETL	geolocation.csv	dim_geolocation

The ETL pipeline successfully transformed raw e-commerce datasets into a structured analytical data warehouse consisting of **6 dimension tables and 4 fact tables**. This process ensured data quality, consistency, and readiness for SQL analytics, Power BI dashboards, and AWS cloud deployment.

11. MySQL Database Implementation

Overview

MySQL was used as the central Data Warehouse database for storing transformed e-commerce data. After completing the ETL process, dimension and fact tables were created in MySQL and populated with cleaned datasets. The database serves as the foundation for SQL analytics, business intelligence reporting, and Power BI dashboard development.

11.1 Database Design

Description

The database was designed using a **Star Schema Architecture**, consisting of dimension tables and fact tables. This design improves query performance, simplifies reporting, and supports scalable analytical processing.

Database Name

brazilian_ecommerce_dw

Schema Components

Dimension Tables

Table Name	Purpose
dim_customers	Customer information
dim_products	Product information
dim_sellers	Seller information
dim_dates	Date dimension
dim_geolocation	Geographic information
dim_payment_types	Payment method information

Fact Tables

Table Name	Purpose
fact_orders	Order transactions
fact_order_items	Product-level sales transactions
fact_payments	Payment transactions
fact_reviews	Customer review transactions

Entity Relationship Overview

The database relationships were established using primary keys and foreign keys to maintain referential integrity between fact and dimension tables.

Key Relationships

Fact Table	Related Dimension
fact_orders	dim_customers
fact_orders	dim_dates
fact_order_items	dim_products
fact_order_items	dim_sellers
fact_payments	dim_payment_types
fact_reviews	dim_dates
dim_customers	dim_geolocation
dim_sellers	dim_geolocation

11.2 Table Creation

Description

After designing the schema, SQL scripts were used to create all dimension and fact tables in MySQL.

Database Objects Created

Object Type	Count
Database	1
Dimension Tables	6
Fact Tables	4
Total Tables	10

11.3 Data Loading

Description

After table creation, cleaned and transformed datasets were loaded into MySQL using Python, Pandas, and MySQL Connector.

Loading Process

Step 1: Establish Database Connection

```
import mysql.connector

connection = mysql.connector.connect(
    host="localhost",
    user="root",
    password="Sunil123",
    database="brazilian_ecommerce_dw"
)
```

Step 2: Read Cleaned Dataset

```
import pandas as pd

customers = pd.read_csv("cleaned_customers.csv")
```

Step 3: Load Data into MySQL

```
from sqlalchemy import create_engine

engine = create_engine(
    "mysql+pymysql://root:password@localhost/brazilian_ecommerce_dw"
)
```

load_dim_customers.csv

```
import pandas as pd

from sqlalchemy import create_engine


# MYSQL CONNECTION

username = "root"

password = "Sunil123"

host = "localhost"

database = "brazilian_ecommerce_dw"


engine = create_engine(

    f"mysql+pymysql://{username}:{password}@{host}/{database}"

)
```

```
# LOAD CSV
df = pd.read_csv(
    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\dim_customers.csv"
)
print(df.head())
# LOAD TO MYSQL
df.to_sql(
    name="dim_customers",
    con=engine,
    if_exists="replace",
    index=False
)
print("dim_customers LOADED SUCCESSFULLY")

# py load_dim_customers.py
```

load_dim_customers.csv

```
import pandas as pd
from sqlalchemy import create_engine

# MYSQL CONNECTION
username = "root"
password = "Sunil123"
host = "localhost"
database = "brazilian_ecommerce_dw"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}/{database}"
)
# LOAD CSV
```

```
df = pd.read_csv(
    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\dim_customers.csv"
)
print(df.head())

# LOAD TO MYSQL
df.to_sql(
    name="dim_customers",
    con=engine,
    if_exists="replace",
    index=False
)
print("dim_customers LOADED SUCCESSFULLY")

# py load_dim_customers.py
```

Load_dim_dates.csv

```
import pandas as pd
from sqlalchemy import create_engine

# MYSQL CONNECTION
username = "root"
password = "Sunil123"
host = "localhost"
database = "brazilian_ecommerce_dw"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}/{database}"
)
print("MYSQL CONNECTION SUCCESSFUL")
```

```
# LOAD CSV
df = pd.read_csv(
    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\dim_dates.csv"
)
print("\n FIRST 5 ROWS")
print(df.head())
print("\n DATA INFO")
print(df.info())

# LOAD TO MYSQL
df.to_sql(
    name="dim_dates",
    con=engine,
    if_exists="replace",
    index=False
)
print("\n dim_dates LOADED SUCCESSFULLY")

# py load_dim_dates.py
```

load_dim_geolocation.csv

```
import pandas as pd
from sqlalchemy import create_engine

# MYSQL CONNECTION
username = "root"
password = "Sunil123"
host = "localhost"
```

```
database = "brazilian_ecommerce_dw"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}/{database}"
)
print("MYSQL CONNECTION SUCCESSFUL")

# LOAD CSV
df = pd.read_csv(
    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\dim_geolocation.csv"
)

print("\n FIRST 5 ROWS")
print(df.head())
print("\n DATA INFO")
print(df.info())
# LOAD TO MYSQL
df.to_sql(
    name="dim_geolocation",
    con=engine,
    if_exists="replace",
    index=False
)
print("\n dim_geolocation LOADED SUCCESSFULLY")

# py load_dim_geolocation.py
```

load_dim_payment_types.csv

```
import pandas as pd
from sqlalchemy import create_engine

# MYSQL CONNECTION
username = "root"
password = "Sunil123"
host = "localhost"
database = "brazilian_ecommerce_dw"
engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}/{database}"
)
print("MYSQL CONNECTION SUCCESSFUL")
# LOAD CSV
df = pd.read_csv(
    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\dim_payment_types.csv"
)
print("\n FIRST 5 ROWS")
print(df.head())
print("\n DATA INFO")
print(df.info())

# LOAD TO MYSQL
df.to_sql(
    name="dim_payment_types",
    con=engine,
    if_exists="replace",
    index=False
)
```

```
print("\n dim_payment_types LOADED SUCCESSFULLY")


# py load_dim_payment_types.py
```

load_dim_products.csv

```
import pandas as pd
from sqlalchemy import create_engine

# MYSQL CONNECTION
username = "root"
password = "Sunil123"
host = "localhost"
database = "brazilian_ecommerce_dw"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}/{database}"
)
print("MYSQL CONNECTION SUCCESSFUL")

# LOAD CSV
df = pd.read_csv(
    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING
PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\dim_products.csv"
)
print("\n FIRST 5 ROWS")
print(df.head())
print("\n DATA INFO")
print(df.info())

# LOAD TO MYSQL
```

```
df.to_sql(
    name="dim_products",
    con=engine,
    if_exists="replace",
    index=False
)
print("\n dim_products LOADED SUCCESSFULLY")


# py load_dim_products.py
```

load_dim_sellers.csv

```
import pandas as pd
from sqlalchemy import create_engine

# MYSQL CONNECTION
username = "root"
password = "Sunil123"
host = "localhost"
database = "brazilian_ecommerce_dw"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}/{database}"
)

print("MYSQL CONNECTION SUCCESSFUL")

# LOAD CSV
df = pd.read_csv(
    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\dim_sellers.csv"
```

```
)
print("\n FIRST 5 ROWS")
print(df.head())
print("\n DATA INFO")
print(df.info())

# LOAD TO MYSQL
df.to_sql(
    name="dim_sellers",
    con=engine,
    if_exists="replace",
    index=False
)
print("\n dim_sellers LOADED SUCCESSFULLY")

# py load_dim_sellers.py
```

load_fact_order_items.py

```
import pandas as pd
from sqlalchemy import create_engine

# MYSQL CONNECTION
username = "root"
password = "Sunil123"
host = "localhost"
database = "brazilian_ecommerce_dw"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}/{database}"
)
```

```
print("MYSQL CONNECTION SUCCESSFUL")

# LOAD CSV
df = pd.read_csv(
    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\fact_order_items.csv"
)
print("\n FIRST 5 ROWS")
print(df.head())
print("\n DATA INFO")
print(df.info())

# LOAD TO MYSQL
df.to_sql(
    name="fact_order_items",
    con=engine,
    if_exists="replace",
    index=False
)
print("\n fact_order_items LOADED SUCCESSFULLY")

# py load_fact_order_items.py
```

load_fact_orders.csv

```
import pandas as pd
from sqlalchemy import create_engine

# MYSQL CONNECTION
username = "root"
password = "Sunil123"
```

```
host = "localhost"
database = "brazilian_ecommerce_dw"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}/{database}"
)
print("MYSQL CONNECTION SUCCESSFUL")

# LOAD CSV
df = pd.read_csv(
    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\fact_orders.csv"
)

print("\n FIRST 5 ROWS")
print(df.head())
print("\n DATA INFO")
print(df.info())

# LOAD TO MYSQL
df.to_sql(
    name="fact_orders",
    con=engine,
    if_exists="replace",
    index=False
)
print("\n fact_orders LOADED SUCCESSFULLY")


# py load_fact_orders.py
```

load_fact_payments.csv

```
import pandas as pd

from sqlalchemy import create_engine


# MYSQL CONNECTION

username = "root"

password = "Sunil123"

host = "localhost"

database = "brazilian_ecommerce_dw"


engine = create_engine(

    f"mysql+pymysql://{username}:{password}@{host}/{database}"

)


print("MYSQL CONNECTION SUCCESSFUL")


# LOAD CSV

df = pd.read_csv(

    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING 
PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\fact_payments.csv"

)

print("\n FIRST 5 ROWS")

print(df.head())

print("\n DATA INFO")

print(df.info())


# LOAD TO MYSQL

df.to_sql(

    name="fact_payments",

    con=engine,
```

```
    if_exists="replace",
    index=False
)
print("\n fact_payments LOADED SUCCESSFULLY")

# py load_fact_payments.py
```

load_fact_reviews.csv

```
import pandas as pd
from sqlalchemy import create_engine

# MYSQL CONNECTION
username = "root"
password = "Sunil123"
host = "localhost"
database = "brazilian_ecommerce_dw"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}/{database}"
)
print("MYSQL CONNECTION SUCCESSFUL")

# LOAD CSV
df = pd.read_csv(
    r"C:\Users\manip\Downloads\HYD IN\PYTHON IMP\DATA ENGINEERING PROJECT\Brazilian_Ecommerce_Data\Data\Analytics_Tables\fact_reviews.csv"
)

print("\n FIRST 5 ROWS")
print(df.head())
```

```
print("\n DATA INFO")
print(df.info())

# LOAD TO MYSQL
df.to_sql(
    name="fact_reviews",
    con=engine,
    if_exists="replace",
    index=False
)
print("\n fact_reviews LOADED SUCCESSFULLY")

# py load_fact_reviews.py
```

Loading Sequence

To maintain data integrity, tables were loaded in the following order:

Dimension Tables

1. dim_customers
2. dim_products
3. dim_sellers
4. dim_dates
5. dim_geolocation
6. dim_payment_types

Fact Tables

7. fact_orders
8. fact_order_items
9. fact_payments
10. fact_reviews

Data Validation

After loading, validation checks were performed:

Row Count Validation

```
SELECT COUNT(*) FROM dim_customers;
SELECT COUNT(*) FROM fact_orders;
```

Null Value Validation

```
SELECT *
FROM dim_customers
WHERE customer_id IS NULL;
```

Relationship Validation

```
SELECT COUNT(*)
FROM fact_orders fo
LEFT JOIN dim_customers dc
ON fo.customer_key = dc.customer_key;
```

Implementation Results

Component	Status
Database Creation	Completed
Table Creation	Completed
Dimension Loading	Completed
Fact Loading	Completed
Data Validation	Completed
SQL Analytics Ready	Completed
Power BI Integration Ready	Completed

12SQL Analytics Layer

The SQL Analytics Layer was developed on top of the MySQL Data Warehouse to transform transactional data into meaningful business insights. A collection of analytical SQL queries and business intelligence views were created to support KPI reporting, business analysis, and Power BI dashboard development.

The analytics layer covers sales, customers, products, sellers, payments, logistics, geolocation, customer satisfaction, and advanced business analytics.

12.1 KPI Analysis

Key Performance Indicators (KPIs) were developed to monitor overall business performance, including Total Revenue, Total Orders, Total Customers, Average Order Value, and Average Review Score.

12.3 Sales Analysis

The Sales Analysis module provides insights into revenue trends and product sales performance.

Queries Developed:
• daily_revenue.sql
• monthly_revenue_trend.sql
• top_revenue_products.sql

12.4 Customer Analysis

The Customer Analysis module evaluates customer distribution and spending behavior.

Queries Developed:
• total_customers.sql
• customers_by_state.sql
• top_10_customers_by_spending.sql

12.5 Product Analysis

The Product Analysis module measures product and category performance.

Queries Developed:
• product_revenue_analysis.sql
• top_product_categories.sql
• highest_freight_cost_products.sql

12.6 Seller Analysis

The Seller Analysis module evaluates seller contribution and marketplace performance.

Queries Developed:
• sellers_by_state.sql
• top_sellers_by_revenue.sql

12.7 Delivery Analysis

The Delivery Analysis module measures logistics efficiency and delivery performance.

Queries Developed:
• average_delivery_time.sql
• orders_delivered_late.sql

12.8 Payment Analysis

The Payment Analysis module analyzes payment behavior and transaction patterns.

Queries Developed:
• payment_type_usage.sql
• revenue_by_payment_type.sql

12.9 Review Analysis

The Review Analysis module measures customer satisfaction using review ratings.

Queries Developed:
• average_review_score.sql
• review_score_distribution.sql

12.10 Geolocation Analysis

The Geolocation Analysis module provides geographic insights into customers and revenue.

Queries Developed:
• customers_by_city.sql
• revenue_by_state.sql

12.11 Window Functions

Advanced SQL window functions were implemented for ranking and cumulative analysis.

Queries Developed:
• rank_sellers_by_revenue.sql
• running_revenue_total.sql

12.12 Advanced Analytics

Advanced business metrics were developed to support strategic decision-making.

Queries Developed:
• customer_lifetime_value.sql
• repeat_customers.sql

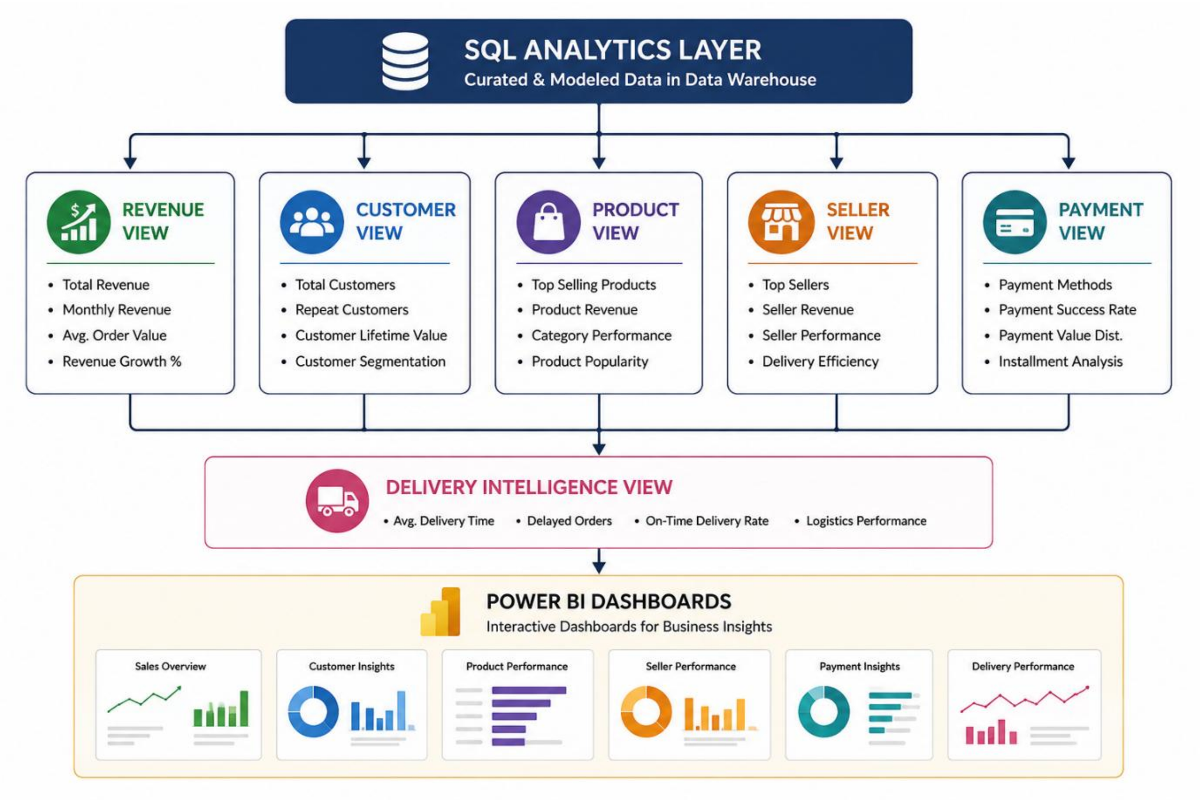
12.13 Business Intelligence Views

The following business views were created for Power BI reporting and executive dashboards:

• vw_executive_sales_summary
• vw_customer_360

• vw_product_performance
• vw_delivery_intelligence
• vw_payment_intelligence
• vw_geographic_revenue_intelligence
• vw_review_satisfaction_intelligence

The SQL Analytics Layer consists of analytical queries, advanced SQL techniques, and business intelligence views that convert warehouse data into actionable insights. These assets serve as the foundation for KPI reporting, business analysis, and Power BI dashboard development.



13. Power BI Dashboard Development

Power BI was used to transform the analytical data warehouse into an interactive Business Intelligence solution. Multiple dashboards were developed to visualize key business metrics, monitor operational performance, and support data-driven decision-making.

The dashboards are connected to the MySQL Data Warehouse and SQL Analytics Layer, providing stakeholders with real-time insights into sales performance, customer behavior, product trends, and logistics efficiency.

13.1 Executive Dashboard

Purpose

The Executive Dashboard provides a high-level overview of overall business performance through key performance indicators (KPIs) and summary visualizations.

KPIs Displayed

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value
- Average Review Score

Visualizations

- KPI Cards
- Revenue Trend Analysis
- Revenue by State
- Top Product Categories
- Payment Type Distribution

Business Value

- Executive performance monitoring
- Revenue tracking
- Strategic decision support
- Business growth analysis

13.2 Sales Analysis Dashboard

Purpose

The Sales Analysis Dashboard focuses on revenue performance, sales trends, and product sales analysis.

Key Metrics

- Total Revenue
- Monthly Revenue Growth
- Daily Revenue
- Product Revenue
- Average Order Value

Visualizations

- Monthly Revenue Trend Line Chart
- Daily Revenue Analysis
- Top Revenue Products
- Revenue by Product Category
- Revenue by State

Business Insights

- Identify peak sales periods
- Monitor revenue growth trends
- Analyze top-performing products
- Evaluate regional sales performance

Business Value

- Sales forecasting
- Revenue optimization
- Product strategy planning

13.3 Customer Analysis Dashboard

Purpose

The Customer Analysis Dashboard provides insights into customer demographics, spending behavior, and geographic distribution.

Key Metrics

- Total Customers
- Customer Revenue
- Average Customer Spend
- Repeat Customers

Visualizations

- Customers by State
- Customers by City
- Top 10 Customers by Spending
- Customer Revenue Distribution
- Customer Segmentation Analysis

Business Insights

- High-value customer identification
- Customer geographic distribution
- Customer purchasing patterns
- Customer retention opportunities

Business Value

- Customer relationship management
- Marketing campaign optimization
- Customer retention improvement

13.4 Product Analysis Dashboard

Purpose

The Product Analysis Dashboard evaluates product performance and category-level sales trends.

Key Metrics

- Product Revenue
- Units Sold
- Average Product Price
- Category Revenue

Visualizations

- Top Product Categories
- Product Revenue Analysis
- Top Revenue Products
- Highest Freight Cost Products
- Category Performance Chart

Business Insights

- Best-selling products
- Product profitability
- Category growth opportunities
- Product demand trends

Business Value

- Inventory planning
- Product portfolio optimization
- Category management

13.5 Logistics Dashboard

Purpose

The Logistics Dashboard measures delivery performance and operational efficiency.

Key Metrics

- Average Delivery Time
- Delayed Orders
- On-Time Deliveries
- Freight Cost Analysis

Visualizations

- Average Delivery Time Chart

- Orders Delivered Late Analysis
- Delivery Performance Trend
- Freight Cost Distribution
- Regional Delivery Performance

Business Insights

- Delivery bottlenecks
- Logistics efficiency trends
- Shipping cost optimization opportunities
- Service level performance

Business Value

- Improved customer satisfaction
- Reduced delivery delays
- Enhanced logistics operations
- Operational cost optimization

Dashboard Summary

Dashboard	Purpose
Executive Dashboard	Business Performance Monitoring
Sales Analysis Dashboard	Revenue & Sales Analysis
Customer Analysis Dashboard	Customer Behavior Analysis
Product Analysis Dashboard	Product Performance Analysis
Logistics Dashboard	Delivery & Logistics Analysis

The Power BI Dashboard Development phase successfully transformed the analytical data warehouse into an interactive reporting solution. The Executive, Sales, Customer, Product, and Logistics dashboards provide comprehensive visibility into business operations, enabling stakeholders to monitor KPIs, identify trends, and make informed data-driven decisions. These dashboards serve as the final presentation layer of the Brazilian E-Commerce Analytics Platform and demonstrate the practical application of Business Intelligence and Data Visualization techniques.

14. DAX Measures & KPIs

DAX (Data Analysis Expressions) measures were created in Power BI to calculate key business metrics and KPIs. These measures enable dynamic reporting, advanced calculations, trend analysis, customer segmentation, revenue analysis, seller performance evaluation, logistics monitoring, and customer satisfaction measurement.

A total of **35+ DAX measures** were developed to support the Executive, Sales, Customer, Product, and Logistics dashboards.

14.1 Revenue Measures

Total Revenue

Calculates total revenue generated from all completed transactions.

Total Revenue =
SUM(fact_payments[payment_value])

Product Revenue

Calculates revenue generated from product sales.

Product Revenue =
SUM(fact_order_items[price])

Revenue by State

Calculates revenue contribution by customer state.

Revenue by City

Calculates revenue contribution by customer city.

Revenue by Payment Type

Calculates revenue generated by each payment method.

Revenue Growth %

Measures revenue growth over time.

Running Revenue Total

Calculates cumulative revenue trends.

14.2 Order Measures

Total Orders

Calculates total number of orders.

```
Total Orders =
DISTINCTCOUNT(fact_orders[order_id])
```

Average Order Value

Calculates average revenue per order.

```
Average Order Value =
DIVIDE(
    [Total Revenue],
    [Total Orders]
)
```

Shipped Orders

Calculates total shipped orders.

Total Delivered Orders

Calculates total successfully delivered orders.

Cancelled Orders

Calculates total cancelled orders.

14.3 Customer Measures

Total Customers

Calculates total unique customers.

```
Total Customers =
DISTINCTCOUNT(dim_customers[customer_unique_id])
```

Orders Per Customer

Measures average orders placed per customer.

Average Revenue Per Customer

Calculates average customer spending.

Customer Lifetime Value

Measures the total revenue generated by a customer during their relationship with the business.

14.4 Product Measures

Average Product Price

Calculates average product selling price.

Highest Product Price

Returns the highest product price.

Lowest Product Price

Returns the lowest product price.

Median Product Price

Calculates median product price.

Top Product Revenue

Measures revenue generated by top-performing products.

14.5 Seller Measures

Total Sellers

Calculates total sellers on the platform.

Seller Revenue

Calculates revenue generated by sellers.

Seller Orders

Calculates total orders fulfilled by sellers.

Average Revenue Per Seller

Calculates average seller revenue.

Seller Rank

Ranks sellers based on revenue performance.

14.6 Payment Measures

Average Installments

Calculates average number of payment installments.

Payment Type Usage

Measures usage frequency of payment methods.

14.7 Logistics Measures

Average Delivery Days

Calculates average delivery duration.

Total Freight Cost

Calculates total shipping cost.

Freight % of Revenue

Measures freight cost as a percentage of revenue.

Late Deliveries

Calculates number of delayed deliveries.

14.8 Customer Satisfaction Measures

Average Review Score

Calculates average customer review rating.

Average Review Score =
AVERAGE(fact_reviews[review_score])

Total Reviews

Calculates total customer reviews.

5 Star Reviews

Calculates total reviews with a score of 5.

14.9 DAX Measures Summary

Category	Measures
Revenue Analysis	7
Order Analysis	5
Customer Analysis	4
Product Analysis	5
Seller Analysis	5
Payment Analysis	2
Logistics Analysis	4
Review Analysis	3

Total Measures Developed 35+

Business Value

The DAX measures provide:

- Real-time KPI monitoring
- Revenue trend analysis
- Customer behavior insights
- Product performance evaluation
- Seller ranking and performance tracking
- Logistics efficiency monitoring
- Customer satisfaction measurement
- Executive-level business reporting

The DAX measures form the analytical foundation of the Power BI solution by transforming raw warehouse data into meaningful business metrics. These measures enable dynamic calculations, advanced analytics, and interactive dashboard reporting, allowing stakeholders to monitor performance and make data-driven decisions across all areas of the e-commerce business.

15.Business Insights & Findings

The Business Insights & Findings section summarizes the key observations discovered through SQL Analytics, Power BI dashboards, and DAX-based KPI analysis. These insights help stakeholders understand business performance, customer behavior, product demand, seller contribution, payment preferences, and logistics efficiency.

15.1 Revenue Insights

Key Findings

- Revenue is concentrated in a small number of high-performing states and cities.
- Certain regions contribute significantly more revenue than others.
- Monthly revenue trends indicate seasonal fluctuations in customer purchasing behavior.
- A small percentage of products generate a large portion of total revenue.

Business Impact

- Focus marketing investments on high-performing regions.
- Identify opportunities for expansion in underperforming markets.
- Improve inventory planning during peak sales periods.

15.2 Customer Insights

Key Findings

- Customer distribution varies significantly across states.
- A relatively small group of customers contributes a substantial share of total revenue.
- Repeat customers generate higher average revenue than one-time customers.
- Customer Lifetime Value (CLV) varies considerably across customer segments.

Business Impact

- Develop loyalty and retention programs for high-value customers.
- Create targeted marketing campaigns based on customer spending behavior.
- Increase focus on repeat customer acquisition.

15.3 Product Insights

Key Findings

- Certain product categories consistently outperform others in revenue generation.
- Top-selling products contribute significantly to overall sales performance.
- High-priced products generate greater revenue despite lower sales volume.
- Some products have disproportionately high freight costs.

Business Impact

- Prioritize inventory management for top-performing categories.
- Optimize pricing strategies for high-demand products.
- Evaluate shipping strategies for products with high freight expenses.

15.4 Seller Insights

Key Findings

- Revenue generation is concentrated among a limited number of sellers.
- Top-ranked sellers contribute a significant percentage of marketplace revenue.
- Seller performance varies considerably by geographic region.
- High-performing sellers maintain consistent order volumes.

Business Impact

- Strengthen partnerships with top-performing sellers.
- Develop seller performance monitoring programs.
- Encourage underperforming sellers through incentive programs.

15.5 Payment Insights

Key Findings

- Certain payment methods dominate customer transactions.
- Customers frequently utilize installment payment options.
- Average payment value varies by payment type.
- Payment preferences differ across customer segments.

Business Impact

- Optimize payment offerings around popular methods.

- Promote installment options to increase order values.
- Improve customer checkout experience.

15.6 Logistics Insights

Key Findings

- Delivery times vary across geographic regions.
- Some orders experience significant delivery delays.
- Freight costs represent a notable percentage of revenue.
- Delivery performance directly impacts customer satisfaction.

Business Impact

- Improve logistics planning and carrier performance.
- Reduce delivery delays to enhance customer experience.
- Optimize shipping costs and fulfillment processes.

15.7 Customer Satisfaction Insights

Key Findings

- Review scores provide a strong indicator of customer satisfaction.
- Orders delivered on time generally receive higher ratings.
- Negative reviews are often associated with delivery delays and service issues.
- A large proportion of customers provide positive ratings.

Business Impact

- Focus on reducing delivery-related issues.
- Improve customer support processes.
- Monitor customer feedback continuously to enhance service quality.

15.8 Geographic Insights

Key Findings

- Revenue and customer concentration are highest in specific states and cities.
- Customer density varies significantly across regions.
- Geographic analysis reveals potential opportunities for market expansion.

Business Impact

- Develop region-specific marketing strategies.
- Expand operations into high-growth areas.
- Improve regional logistics planning.

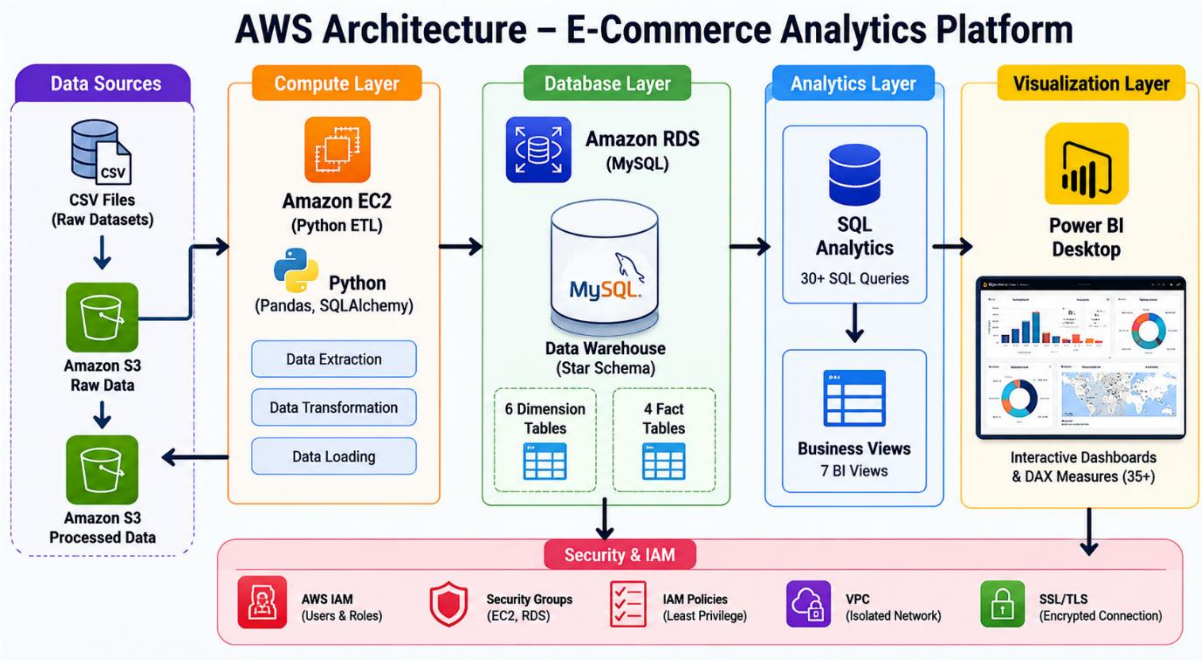
15.9 Executive Summary of Findings

Business Area	Key Insight
Revenue	Revenue is concentrated in a few high-performing regions
Customers	High-value and repeat customers drive significant revenue
Products	A small number of products and categories contribute most sales
Sellers	Top sellers generate a major share of marketplace revenue
Payments	Certain payment methods dominate transactions
Logistics	Delivery performance strongly influences customer experience
Reviews	Customer satisfaction is closely linked to delivery quality
Geography	Significant regional differences exist in revenue and customer distribution

The analysis demonstrates that business performance is driven by a combination of customer behavior, product demand, seller effectiveness, payment preferences, and logistics efficiency. By leveraging insights generated from SQL Analytics, Power BI dashboards, and DAX measures, stakeholders can make informed decisions to improve revenue growth, customer satisfaction, operational efficiency, and overall marketplace performance. These findings highlight the value of a data-driven approach to managing and optimizing e-commerce operations.

16. AWS Cloud Deployment

16.1 AWS Architecture



16.2 EC2 Setup

- Python ETL Processing
- Pandas
- SQLAlchemy
- Data Loading

16.3 RDS Setup

- MySQL Data Warehouse
- Fact Tables
- Dimension Tables

16.4 S3 Storage

- Raw Data Storage
- Processed Data Storage

16.5 Security & IAM

- IAM Users & Roles
- Security Groups
- Access Control

17. Challenges Faced

During the development of the Brazilian E-Commerce Analytics Platform, several technical and analytical challenges were encountered across data processing, data warehousing, SQL analytics, Power BI development, and cloud deployment. Overcoming these challenges provided valuable hands-on experience in real-world data engineering and business intelligence practices.

17.1 Data Quality Issues

Challenge

The raw datasets contained missing values, inconsistent records, and duplicate entries that affected data quality and analysis accuracy.

Solution

- Performed data profiling using Python and Pandas.
- Identified and handled null values.
- Removed duplicate records.
- Standardized data formats and naming conventions.

Outcome

Improved data consistency and reliability for downstream analytics.

17.2 Handling Large Datasets

Challenge

Processing multiple large CSV files containing millions of records required efficient data handling techniques.

Solution

- Used Pandas for efficient data processing.
- Selected only required columns during transformations.
- Optimized memory usage during ETL execution.

Outcome

Successfully processed large datasets with improved performance.

17.3 Data Transformation Complexity

Challenge

Creating relationships between customers, orders, products, sellers, payments, and reviews required careful transformation logic.

Solution

- Designed a structured ETL workflow.
- Applied business rules consistently.
- Created surrogate and foreign key mappings.

Outcome

Established accurate relationships across all warehouse tables.

17.4 Star Schema Design

Challenge

Designing an efficient dimensional model while maintaining data integrity and analytical flexibility.

Solution

- Implemented a Star Schema architecture.
- Separated fact and dimension tables.
- Defined appropriate relationships and keys.

Outcome

Created a scalable and reporting-friendly data warehouse.

17.5 SQL Query Optimization

Challenge

Developing analytical queries involving multiple joins and aggregations resulted in performance concerns.

Solution

- Optimized joins and filtering conditions.
- Created reusable SQL business views.
- Simplified analytical reporting through view creation.

Outcome

Improved query performance and reporting efficiency.

17.6 Power BI Data Modeling

Challenge

Managing relationships between multiple fact and dimension tables in Power BI.

Solution

- Implemented a Star Schema model within Power BI.
- Configured one-to-many relationships.
- Used proper filtering directions.

Outcome

Improved dashboard performance and data consistency.

17.7 DAX Measure Development

Challenge

Creating advanced KPIs such as Customer Lifetime Value, Running Revenue Total, Seller Ranking, and Revenue Growth.

Solution

- Developed reusable DAX measures.
- Applied aggregation and time-intelligence functions.
- Validated calculations against SQL outputs.

Outcome

Created dynamic and accurate business metrics.

17.8 Dashboard Design and Visualization

Challenge

Selecting appropriate visualizations to effectively communicate business insights.

Solution

- Designed dashboard layouts based on business requirements.
- Used KPI cards, charts, maps, slicers, and drill-through functionality.
- Focused on user-friendly visualization design.

Outcome

Developed interactive and intuitive dashboards.

17.9 Business View Creation

Challenge

Converting complex SQL analytics into reusable business intelligence views.

Solution

- Created specialized SQL views for:
  - Revenue Intelligence
  - Customer Intelligence
  - Product Performance
  - Payment Intelligence
  - Delivery Intelligence
  - Review Satisfaction

Outcome

Simplified Power BI integration and business reporting.

17.10 AWS Cloud Deployment Preparation

Challenge

Understanding cloud deployment architecture and integrating AWS services with the analytics platform.

Solution

- Designed AWS deployment architecture.
- Planned integration of Amazon S3, EC2, RDS, and IAM.
- Studied cloud security and access management concepts.

Outcome

Prepared a cloud-ready analytics solution for future deployment.

17.11 End-to-End Integration

Challenge

Ensuring smooth integration between ETL pipelines, MySQL, SQL Analytics, Power BI, and AWS components.

Solution

- Validated data flow at each stage.
- Performed extensive testing and verification.
- Implemented consistent naming and modeling standards.

Outcome

Successfully built a complete end-to-end Data Engineering and Business Intelligence solution.

Summary of Challenges

Area	Challenge
Data Processing	Missing values, duplicates, inconsistencies
ETL Development	Complex transformations and relationships
Data Modeling	Star Schema implementation
SQL Analytics	Query optimization and business views
Power BI	Data modeling and DAX calculations
Dashboard Development	Visualization design and KPI reporting
Cloud Computing	AWS architecture and deployment planning
Integration	End-to-end system connectivity

The challenges encountered throughout this project provided practical experience in solving real-world data engineering and analytics problems. By addressing issues related to data quality, ETL processing, data warehousing, SQL analytics, Power BI development, and AWS deployment, the project successfully evolved into a robust end-to-end analytics platform. These challenges enhanced technical expertise and reinforced best practices in Data Engineering, Business Intelligence, and Cloud Analytics.

18. Future Enhancements

Overview

While the Brazilian E-Commerce Analytics Platform successfully delivers an end-to-end Data Engineering and Business Intelligence solution, several enhancements can be implemented in the future to improve scalability, automation, real-time analytics, and predictive capabilities.

These enhancements will further align the platform with modern industry standards and enterprise-level data engineering practices.

18.1 Workflow Automation Using Apache Airflow

Objective

Automate the ETL pipeline and data refresh processes.

Proposed Enhancements

- Schedule ETL jobs automatically.
- Monitor workflow execution.
- Handle task dependencies.
- Generate failure notifications and alerts.

Benefits

- Reduced manual intervention.
- Improved reliability.
- Better workflow management.

18.2 Real-Time Data Processing

Objective

Enable real-time analytics instead of batch processing.

Proposed Enhancements

- Integrate Apache Kafka for data streaming.
- Process events in near real time.
- Update dashboards dynamically.

Benefits

- Faster decision-making.
- Live KPI monitoring.

- Improved business responsiveness.

18.3 Machine Learning Integration

Objective

Introduce predictive analytics capabilities.

Proposed Enhancements

- Sales forecasting.
- Customer churn prediction.
- Customer segmentation.
- Product recommendation systems.
- Demand forecasting.

Benefits

- Predictive decision-making.
- Improved customer retention.
- Revenue growth opportunities.

18.4 Advanced Power BI Dashboards

Objective

Enhance reporting and visualization capabilities.

Proposed Enhancements

- Drill-through reports.
- Advanced tooltips.
- AI-powered insights.
- Dynamic forecasting visuals.
- Mobile-optimized dashboards.

Benefits

- Better user experience.
- Deeper business analysis.
- Enhanced executive reporting.

18.5 Advanced SQL Analytics

Objective

Expand analytical capabilities.

Proposed Enhancements

- Cohort Analysis.
- RFM (Recency, Frequency, Monetary) Analysis.
- Basket Analysis.
- Customer Retention Analysis.
- Seasonal Trend Analysis.

Benefits

- Deeper customer insights.
- Better marketing strategies.
- Improved business intelligence.

18.6 Data Lake Architecture

Objective

Extend storage architecture beyond a traditional data warehouse.

Proposed Enhancements

- Implement Amazon S3 Data Lake.
- Store structured and unstructured data.
- Support future big data workloads.

Benefits

- Greater scalability.
- Lower storage costs.
- Big data readiness.

18.7 Enterprise Monitoring and Alerting

Objective

Improve operational visibility.

Proposed Enhancements

- AWS CloudWatch monitoring.

- ETL failure alerts.
- Database performance monitoring.
- Dashboard refresh monitoring.

Benefits

- Faster issue resolution.
- Improved system reliability.
- Better operational management.

Future Roadmap Summary

Enhancement Area	Technology
Workflow Automation	Apache Airflow
Cloud Deployment	AWS
Real-Time Analytics	Apache Kafka
Machine Learning	Python, Scikit-Learn
Advanced Dashboards	Power BI
Data Quality Monitoring	Great Expectations
Advanced Analytics	SQL, Python
Data Lake	Amazon S3
CI/CD	GitHub Actions
Monitoring & Alerting	AWS CloudWatch

The current project establishes a strong foundation for data engineering and business intelligence. Future enhancements such as workflow automation, AWS deployment, real-time data processing, machine learning integration, advanced analytics, and enterprise monitoring can transform the platform into a production-ready, enterprise-scale analytics solution capable of supporting large-scale e-commerce operations and strategic decision-making.

19. Conclusion

The **Brazilian E-Commerce Analytics Platform** successfully demonstrates the complete lifecycle of a modern Data Engineering and Business Intelligence solution. The project transformed raw e-commerce datasets into a structured and analytics-ready environment through a systematic ETL process, dimensional data modeling, SQL analytics, and interactive Power BI dashboards.

The implementation began with data extraction, cleaning, and transformation using Python and Pandas. The processed data was then organized into a Star Schema Data Warehouse in MySQL, consisting of dimension and fact tables optimized for analytical reporting. A comprehensive SQL Analytics Layer was developed to generate business-focused insights across sales, customers, products, sellers, payments, logistics, reviews, and geographic performance.

Power BI dashboards were created to visualize key business metrics and provide stakeholders with an intuitive platform for monitoring performance and making data-driven decisions. Advanced DAX measures enabled dynamic KPI calculations, trend analysis, customer insights, and operational reporting.

The project also explored cloud deployment architecture using AWS services such as Amazon S3, EC2, RDS, and IAM, preparing the solution for scalable and secure enterprise-level deployment.

Key Achievements

- Developed an end-to-end ETL pipeline using Python and Pandas.
- Built a Star Schema Data Warehouse in MySQL.
- Created dimension and fact tables for analytical processing.
- Developed SQL analytics queries and business intelligence views.
- Designed interactive Power BI dashboards with KPI monitoring.
- Implemented DAX measures for advanced business analysis.
- Designed a cloud-ready AWS deployment architecture.
- Generated actionable business insights from e-commerce data.

Project Impact

The platform enables organizations to:

- Monitor revenue and sales performance.
- Analyze customer purchasing behavior.
- Evaluate product and seller performance.
- Track payment trends and customer preferences.
- Measure delivery efficiency and logistics performance.

- Monitor customer satisfaction through review analysis.
- Support strategic business decisions using data-driven insights.

Final Outcome

This project successfully integrates Data Engineering, Data Warehousing, SQL Analytics, Business Intelligence, and Cloud Computing concepts into a single real-world solution. It demonstrates practical industry-level skills and provides a strong foundation for future enhancements such as workflow automation, real-time analytics, machine learning, and enterprise cloud deployment.

Overall, the Brazilian E-Commerce Analytics Platform showcases the ability to design, build, and deploy a complete analytics ecosystem that transforms raw data into valuable business intelligence, enabling organizations to make informed and strategic decisions.

20. References

The following references, tools, technologies, documentation sources, and datasets were used during the development of the Brazilian E-Commerce Analytics Platform.

20.1 Dataset References

Brazilian E-Commerce Public Dataset by Olist

Source:

- Brazilian E-Commerce Public Dataset (Olist)

Accessed From:

- Kaggle

Reference:

Olist Brazilian E-Commerce Dataset
https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

20.2 Python Documentation

Python Software Foundation

Python Documentation
https://docs.python.org/3/

Used for:

- Data processing
- ETL development
- Data transformation

20.3 Pandas Documentation

Pandas Development Team

Pandas Documentation
https://pandas.pydata.org/docs/

Used for:

- Data cleaning
- Data manipulation
- Data transformation

20.4 MySQL Documentation

Oracle Corporation

MySQL Documentation
https://dev.mysql.com/doc/

Used for:

- Database design
- Table creation
- SQL query development
- Data warehousing

20.5 SQL Reference Documentation

W3Schools SQL Tutorial

https://www.w3schools.com/sql/

Used for:

- SQL query development
- Window functions
- Aggregate functions
- Analytical queries

20.6 Power BI Documentation

Microsoft Learn

Power BI Documentation
https://learn.microsoft.com/power-bi/

Used for:

- Dashboard development
- Data modeling
- Visualization design

20.7 DAX Documentation

Microsoft Learn

DAX Documentation
https://learn.microsoft.com/dax/

Used for:

- KPI development
- Business calculations
- Advanced analytics measures

20.8 AWS Documentation

Amazon Web Services

AWS Documentation
https://docs.aws.amazon.com/

Used for:

- Cloud architecture design
- EC2 deployment
- RDS configuration
- S3 storage architecture
- IAM security implementation

20.9 Git & GitHub Documentation

GitHub Documentation

https://docs.github.com/

Git Documentation

https://git-scm.com/doc

Used for:

- Version control
- Source code management
- Project repository maintenance

20.10 Data Warehousing Reference

Ralph Kimball

The Data Warehouse Toolkit
(Third Edition)

Reference:

Kimball, R., & Ross, M.
The Data Warehouse Toolkit:
The Definitive Guide to Dimensional Modeling.
Wiley Publications.

Used for:

- Star Schema design
- Fact and dimension modeling
- Data warehouse architecture

20.11 Business Intelligence Reference

Microsoft Power BI Learning Resources

Reference:

Microsoft Learn – Power BI Learning Path
https://learn.microsoft.com/training/powerplatform/power-bi/

Used for:

- Dashboard design
- KPI reporting
- Business intelligence implementation

20.12 Development Tools

The following software and tools were used throughout the project:

Tool	Purpose
Python	ETL Development
Pandas	Data Processing
MySQL	Data Warehouse
MySQL Workbench	Database Management
SQL	Analytics Layer
Power BI Desktop	Dashboard Development
AWS	Cloud Deployment
VS Code	Development Environment
Git	Version Control
GitHub	Project Repository

The successful completion of this project was supported by a combination of publicly available datasets, official technology documentation, industry-standard data warehousing practices, and business intelligence resources. These references provided the technical foundation for designing, developing, and documenting the Brazilian E-Commerce Analytics Platform.

21. Appendix

21.1 SQL Scripts

The project includes multiple SQL scripts developed for data analysis and business intelligence reporting.

SQL Analytics Scripts

- daily_revenue.sql
- monthly_revenue_trend.sql
- top_revenue_products.sql
- total_customers.sql
- customers_by_state.sql
- top_10_customers_by_spending.sql
- product_revenue_analysis.sql
- top_product_categories.sql
- highest_freight_cost_products.sql
- sellers_by_state.sql
- top_sellers_by_revenue.sql
- average_delivery_time.sql
- orders_delivered_late.sql
- payment_type_usage.sql
- revenue_by_payment_type.sql
- average_review_score.sql
- review_score_distribution.sql
- customers_by_city.sql
- revenue_by_state.sql
- rank_sellers_by_revenue.sql
- running_revenue_total.sql
- customer_lifetime_value.sql
- repeat_customers.sql
-

21.2 Python ETL Scripts

The following Python scripts were developed for data extraction, cleaning, transformation, and loading.

ETL Scripts

- customer_etl.py
- product_etl.py
- seller_etl.py
- order_etl.py
- payment_etl.py
- review_etl.py
- product_category_etl.py
- order_items_etl.py
- geolocation_etl.py

21.3 Data Warehouse Scripts

The following SQL scripts were used to create the dimensional data warehouse.

Dimension Tables

- create_dim_customers.sql
- create_dim_products.sql
- create_dim_sellers.sql
- create_dim_dates.sql
- create_dim_geolocation.sql
- create_dim_payment_types.sql

Fact Tables

- create_fact_orders.sql
- create_fact_order_items.sql
- create_fact_payments.sql
- create_fact_reviews.sql

Business Views

- vw_executive_sales_summary.sql

- vw_customer_360.sql
- vw_product_performance.sql
- vw_delivery_intelligence.sql
- vw_payment_intelligence.sql
- vw_geographic_revenue_intelligence.sql
- vw_review_satisfaction_intelligence.sql

21.4 Dashboard Screenshots

The appendix includes screenshots of the Power BI dashboards developed during the project.

Dashboard Screenshots

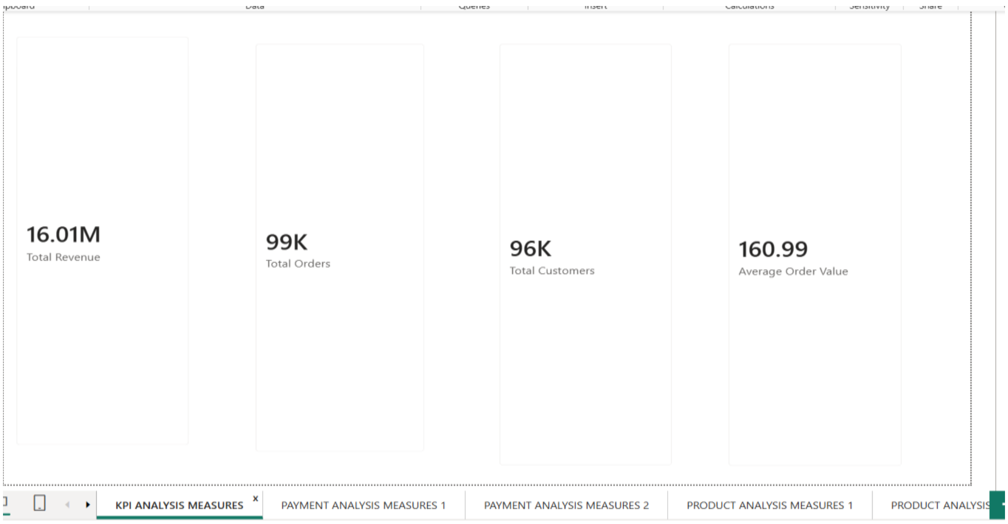


Figure A.1

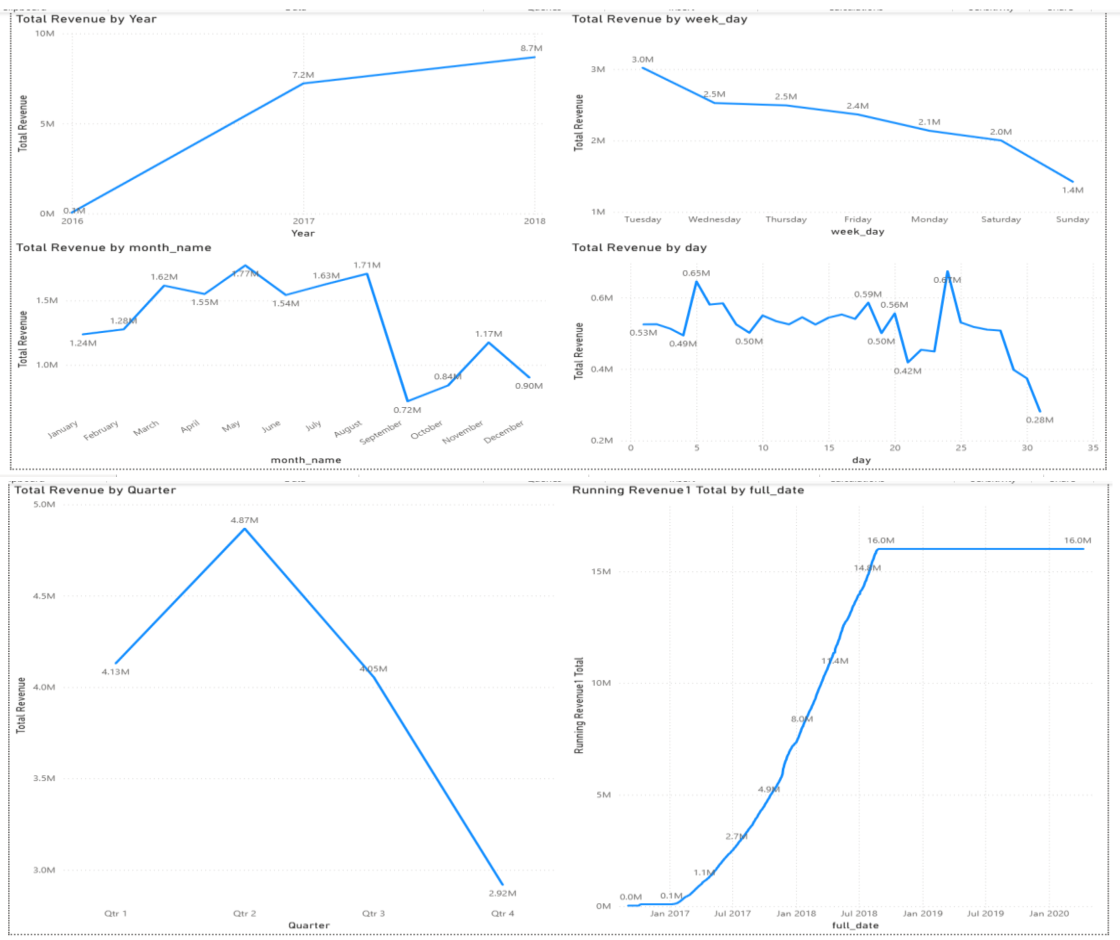


Figure A.2

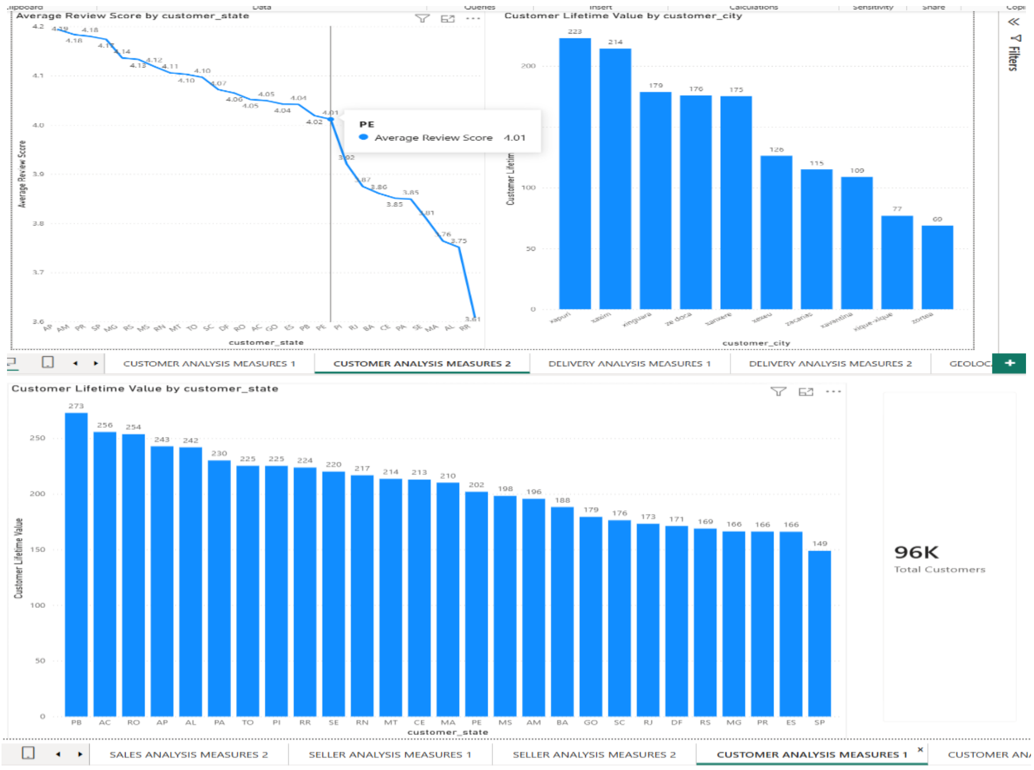


Figure A.3

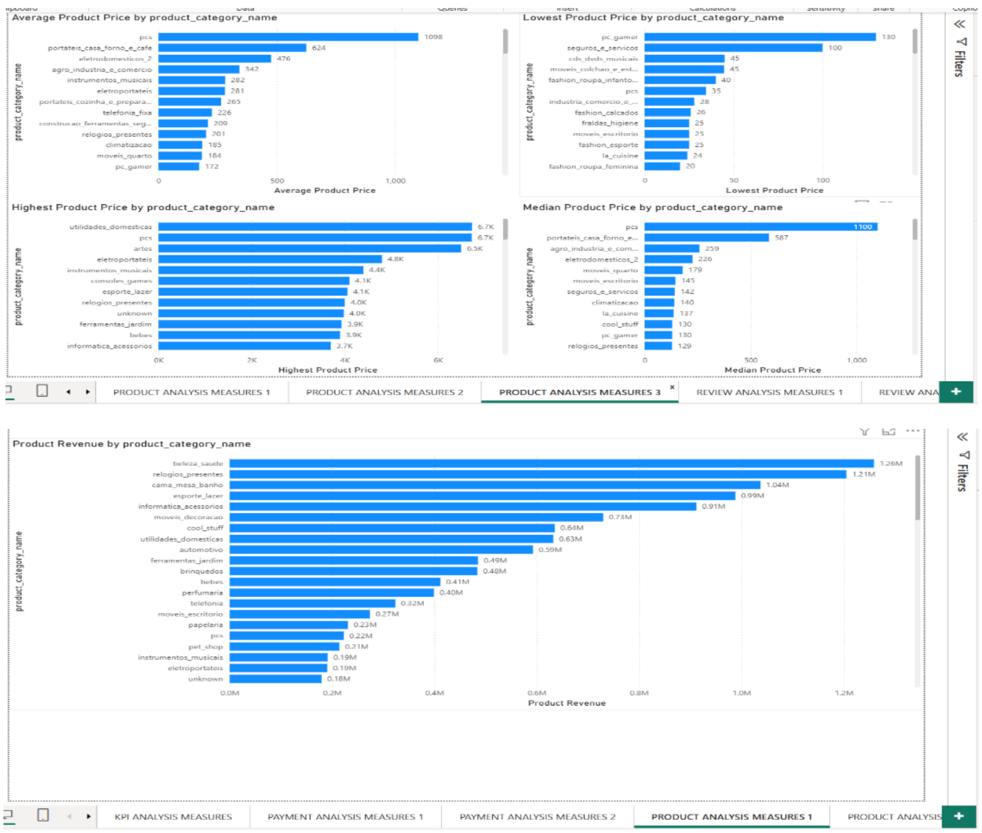


Figure A.4

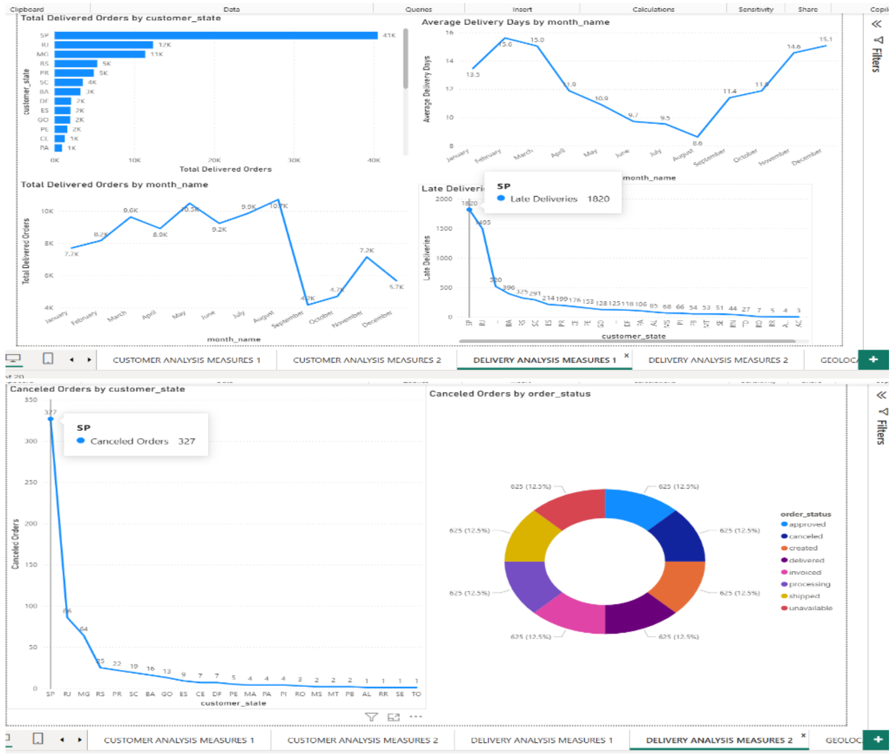
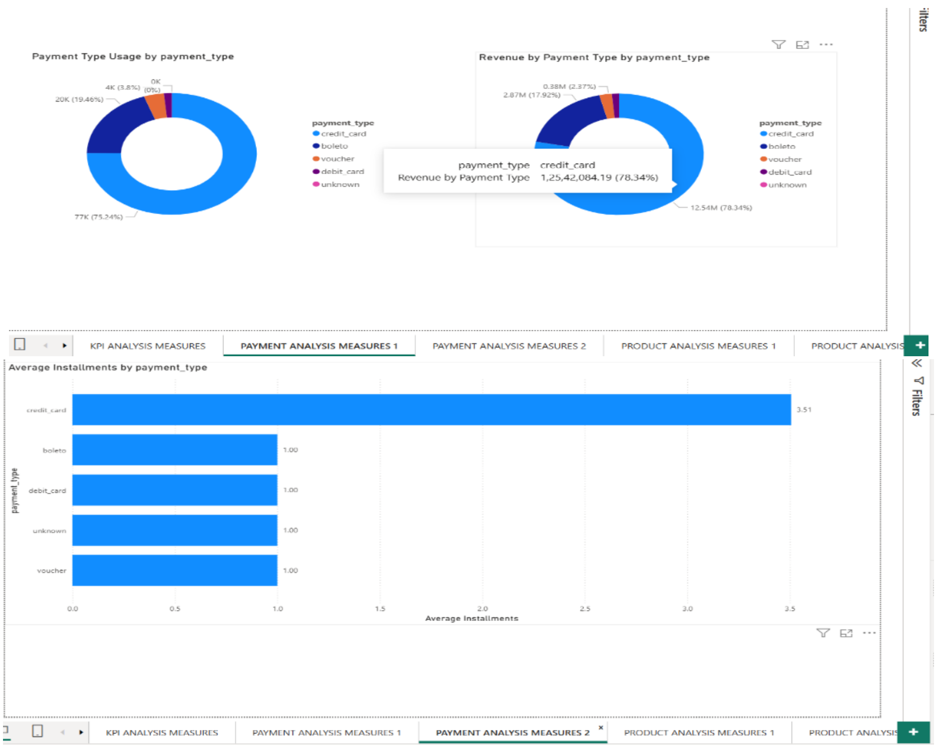


Figure A.5



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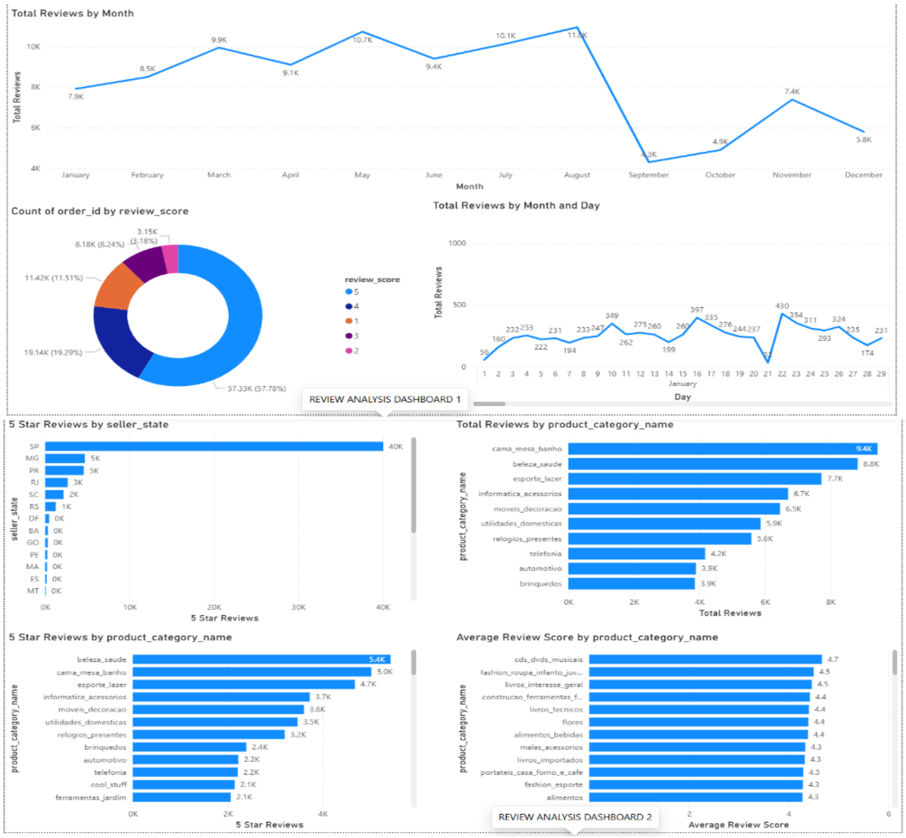

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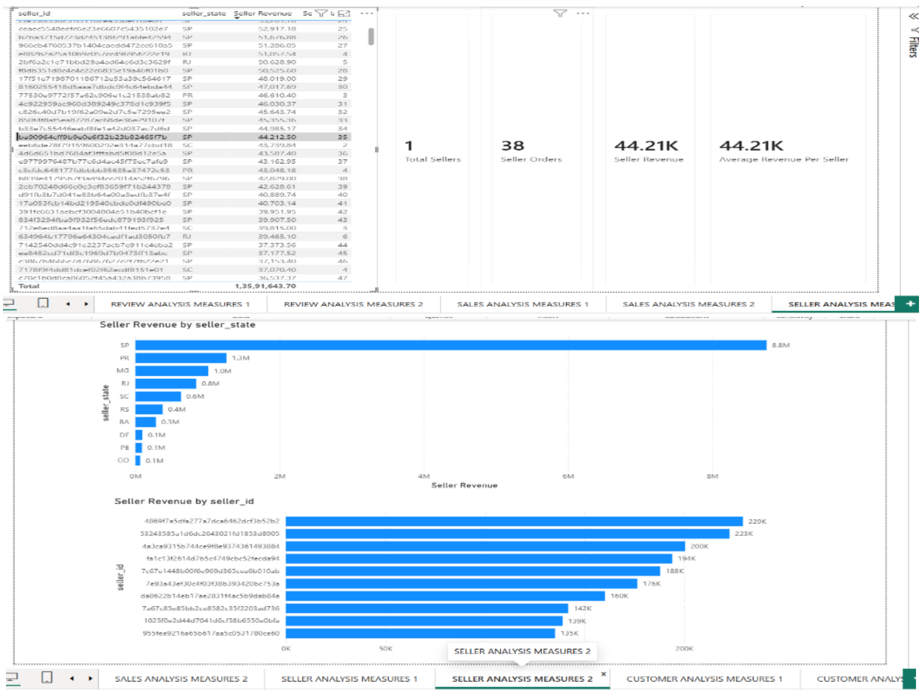

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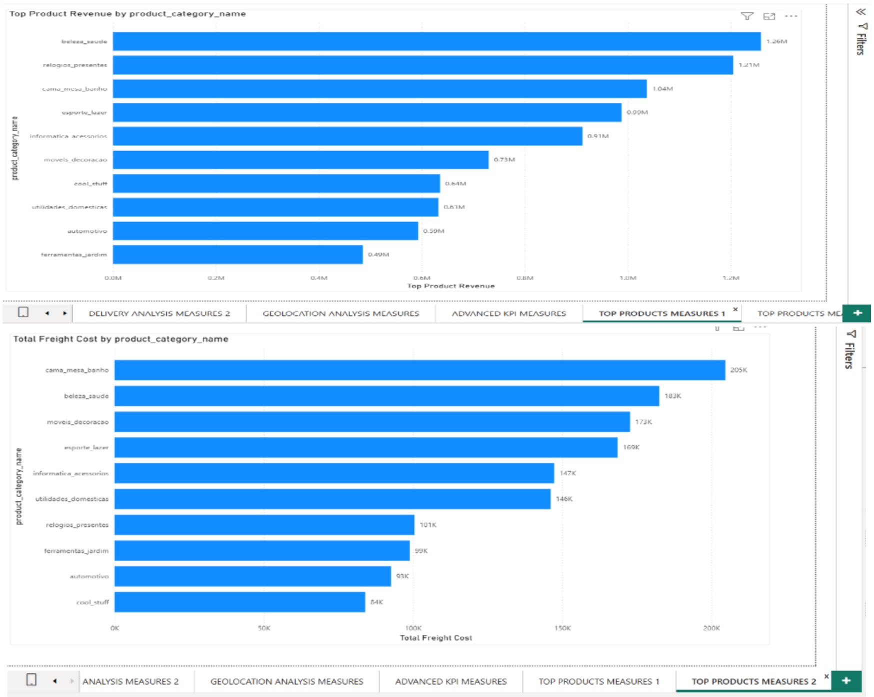

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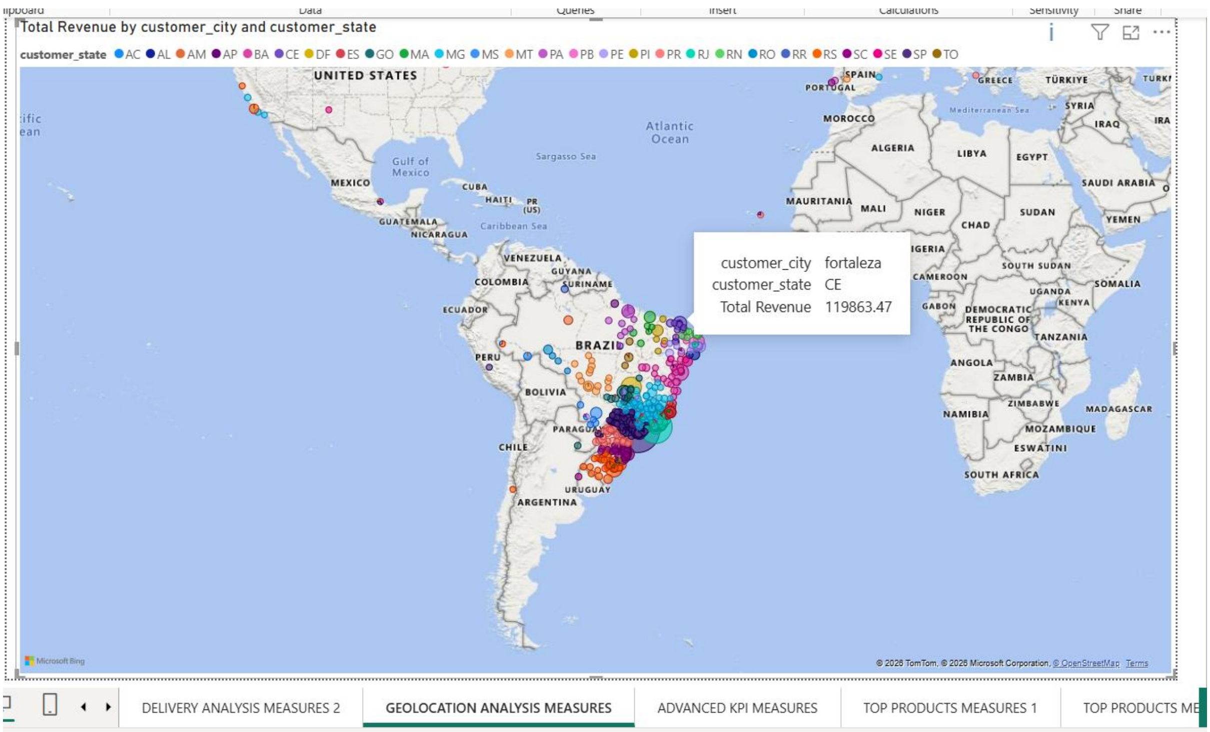

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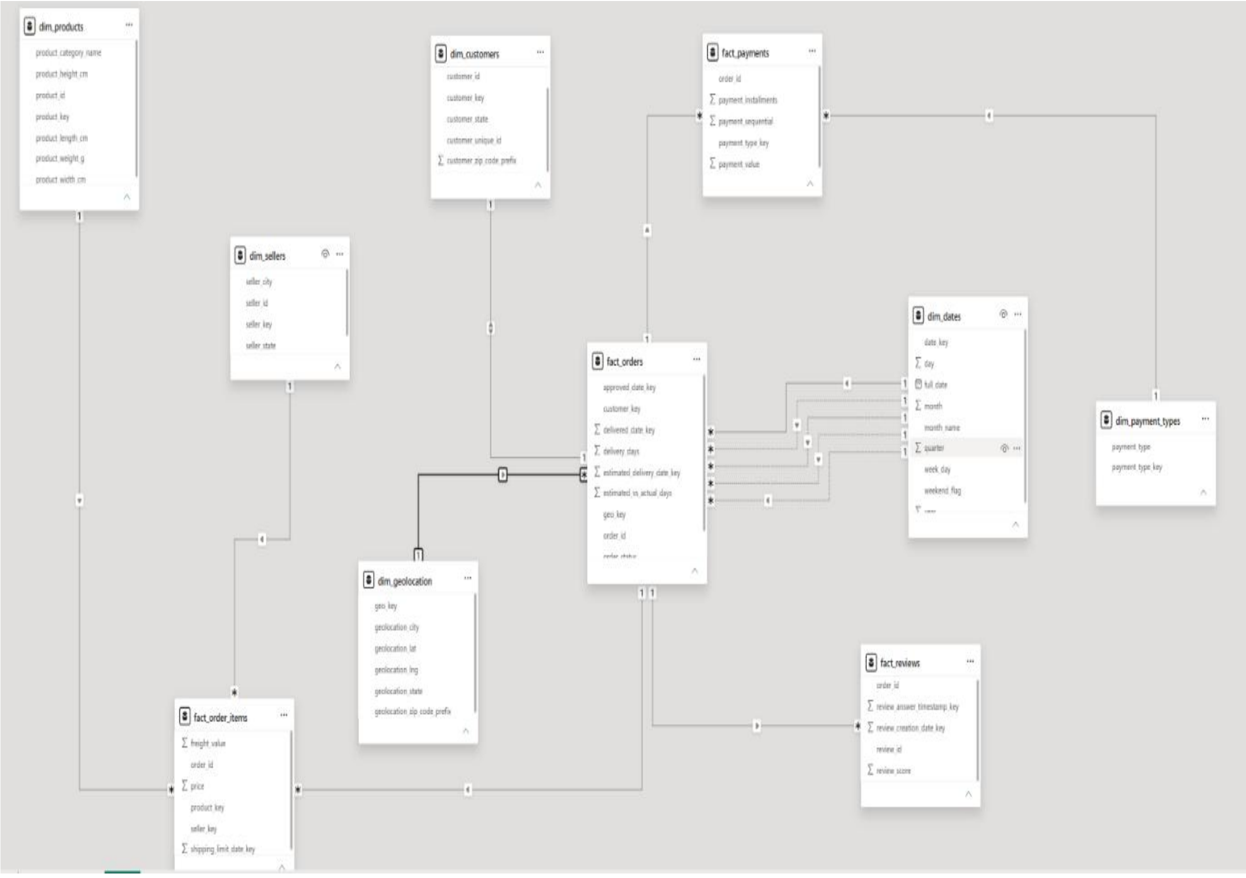
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product height cm
product id
product key
product length cm
product weight g
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customer unique id
customer zip code prefix
fact_payments
order id
payment installments
payment sequential
payment type key
payment value
dim_sellers
seller city
seller id
seller key
seller state
dim_dates
date key
day
full date
month
month name
quarter
week day
weekend flag
fact_orders
approved date key
customer key
delivered date key
delivery days
estimated delivery date key
estimated vs actual days
geo key
order id
dim_payment_types
payment type
payment type key
dim_geolocation
geo key
geolocation city
geolocation lat
geolocation lng
geolocation state
geolocation zip code prefix
fact_reviews
order id
review answer timestamp key
review creation date key
review id
review score
fact_order_items
freight value
order id
price
product key
seller key
shipping limit date key

Figure A.11